ANALOG LAB · LECTURES 25–26 · SWITCHING AND DELAY · TUTORIALS 8 AND 9

Lectures 25–26, made simple

First the whole lecture as one short story. Then each idea, one at a time, with a picture. Then a playbook that tells you which kind of question you are looking at, every question of Tutorials 8 and 9 solved, and ten more worked questions. Each question is solved in the same five steps: **what it says → which theory → plan → solve → check**.

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A · Lecture 25 in 10 minutes

Lectures 23 and 24 were about **DC**: which output voltage you get for each input voltage (the VTC). They never asked *how long* it takes. Lecture 25 asks exactly that: **when the input switches, how long until the output follows?**

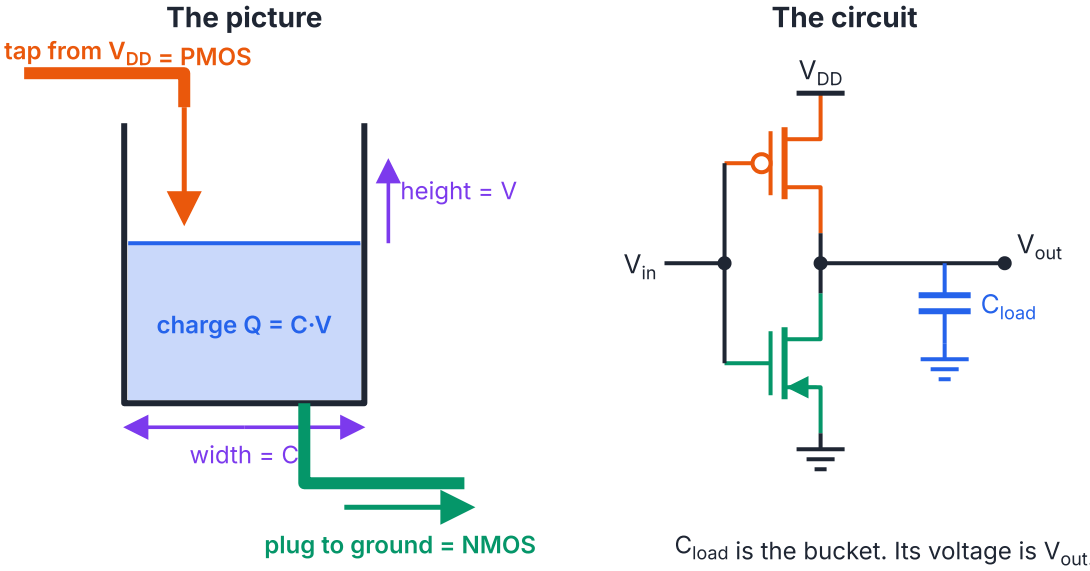


Fig. A.1 The whole lecture in one picture. The output node is a capacitor (a bucket). The PMOS is a tap that fills it from V_{DD}. The NMOS is a plug that empties it into ground. Delay is how long the filling or emptying takes.

1. **The output is a capacitor.** Wires, transistor junctions and the next gate all connect to V_{out}, so V_{out} is the voltage on a capacitor C_{load}. A capacitor's voltage cannot jump. To change it, you must move charge $Q = CV$, and that takes time.

2. **Only one transistor works at a time.** Input goes up → PMOS off, NMOS on → NMOS pulls charge out → output falls. Input goes down → NMOS off, PMOS on → PMOS pushes charge in → output rises.

3. **The one equation:** time = charge to move ÷ current.

$$\tau \approx \frac{C\,\Delta V}{I}$$

Bigger bucket (C) → slower. Bigger tap (I, so bigger k or V_{DD}) → faster. **Every question in this lecture is "find C" or "find I" or both.**

4. **What we measure.** Delay τ_{PHL} (output going High→Low) and τ_{PLH} (Low→High) are measured from the input's 50% point to the output's 50% point. So the output only has to travel **half way**: $\Delta V = V_{DD}/2$. The average delay is $\tau_P = (\tau_{PHL} + \tau_{PLH})/2$.

5. **The difficulty.** The current I is not constant. As the output falls, the NMOS starts in saturation (constant current), then goes linear (current shrinks). So "$C\Delta V/I$" has no single I.

6. **Two ways around it. Method 1 (average current):** work out I at the start and at the 50% point, average them, then use $C\Delta V/I_{avg}$. **Method 2 (exact):** integrate $C\,dV/dt = -I$ in two pieces. The result is one formula you can plug numbers into.

7. **What the formula tells you.** $\tau \propto C/k$. Make the PMOS about 2.5× wider than the NMOS and rise = fall (that is the symmetric inverter of Lecture 24).

8. **Two extras.** A loop of an odd number N of inverters oscillates with $f = 1/(2N\tau_P)$. Every full switch costs energy CV_{DD}^2, so power is $P = CV_{DD}^2 f$.

THE ONE SENTENCE TO REMEMBER

"A gate is slow because its output is a **bucket** and its transistors are a **tap**." Delay = water ÷ flow = $C\Delta V/I$. Everything else is detail about finding C and finding I.

WHAT YOU ALREADY KNOW (FROM LECTURES 23–24) THAT WE REUSE

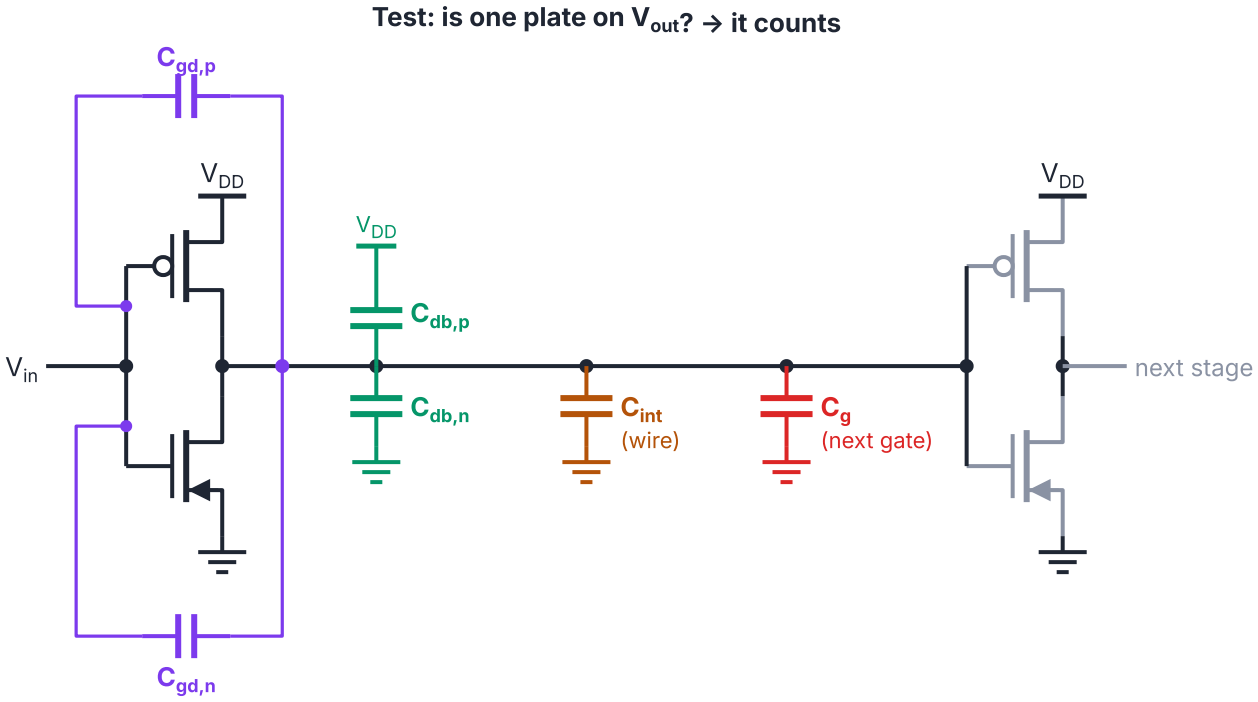
$k_n = \mu_n C_{ox}(W/L)_n$, $k_p = \mu_p C_{ox}(W/L)_p$. With overdrive $u = |V_{GS}| - |V_T|$ and $v = |V_{DS}|$:

$$\text{saturation } (v \geq u):\ I = \tfrac{k}{2}u^2 \qquad\qquad \text{linear } (v < u):\ I = \tfrac{k}{2}\left[2uv - v^2\right]$$

That is all the device physics Lecture 25 needs.

B1 · Where the capacitor comes from

A capacitor is just **two conductors with an insulator between them**. Nobody solders capacitors into a chip. They appear by themselves wherever this shape exists. Around the output of an inverter there are six of them.



All six have one plate on the output node, so every change of the output must fill or empty them.

This inverter's own C_{gs} sits between V_{in} and a rail → it loads the INPUT, not the output.

Fig. B1.1 Inverter 1 drives inverter 2. Count a capacitor if **one of its plates is on** V_{out}. A capacitor to V_{DD} counts just like one to ground, because V_{DD} is also a fixed voltage.

THE LOAD CAPACITANCE (WRITE THIS FIRST IN EVERY DELAY QUESTION)

$$C_{load} = \underbrace{C_{gd,n} + C_{gd,p}}_{\text{overlaps}} + \underbrace{C_{db,n} + C_{db,p}}_{\text{drain junctions}} + \underbrace{C_{int}}_{\text{wire}} + \underbrace{C_g}_{\text{next gate}}$$

TERM	WHAT IT PHYSICALLY IS	FORMULA
C_{gd} (overlap)	The drain spreads a little (L_D) under the gate. Gate over drain with oxide between = small capacitor.	$C_{ox}\,W\,L_D$ or $C_{GDO} \times W$ (if given "per µm of width")
C_{db} (junction)	The n⁺ drain sits in p substrate: a reverse-biased diode. Its depletion region is the insulator.	$WE\,C_j + 2(W+E)\,C_{jsw}$ (area + perimeter, Fig. B1.2)
C_{int}	The metal wire to the next gate.	usually given as a number
C_g (next gate)	Gate oxide of both transistors in the next inverter. **Usually the biggest term.**	$C_{ox}\left[(WL)_n + (WL)_p\right]$ of the next stage

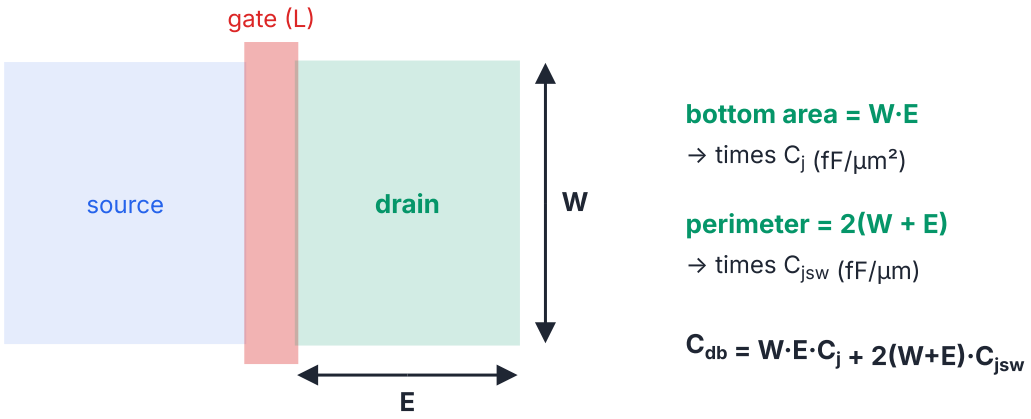


Fig. B1.2 Top view of a drain. The bottom face is $W \times E$ (use C_j, fF/µm²). The side walls run all around: perimeter $2(W + E)$ (use C_{jsw}, fF/µm). Use each transistor's own C_j and C_{jsw}.

THREE CLASSIC MISTAKES

① Adding this inverter's own C_{gs}. It sits between V_{in} and a rail, so it loads the *previous* stage, not this output.
② Forgetting the next stage's gate C_g. It is usually the largest term.
③ Mixing units: C_{ox} in fF/µm² × µm × µm gives fF. Keep everything in fF and µm.

ONLY IF THE QUESTION ASKS "WHY" (THEORY FROM THE LECTURE)

The gate–channel part of the capacitance depends on the region. Off: $C_{GS} = C_{GD} =$ overlap only. Linear: each gets $\frac{1}{2}WLC_{ox}$ + overlap. Saturation: $C_{GS} = \frac{2}{3}WLC_{ox}$ + overlap, but C_{GD} = overlap only, because the channel is pinched off at the drain. For C_{load} the lecture keeps it simple: overlap for C_{gd}, full WLC_{ox} for the next gate. More reverse bias → wider depletion → smaller C_j. "Zero-bias" in a question means: just use the given C_j.

B2 · What happens when the input switches

Put all those capacitors together into one C_{load}. Now Kirchhoff's current law at the output node gives the only equation of the lecture:

$$C_{load}\frac{dV_{out}}{dt} = i_{D,p} - i_{D,n} \qquad \text{(current in from PMOS minus current out through NMOS)}$$

With a sudden (step) input, only one transistor is on, so one of the two currents is zero.

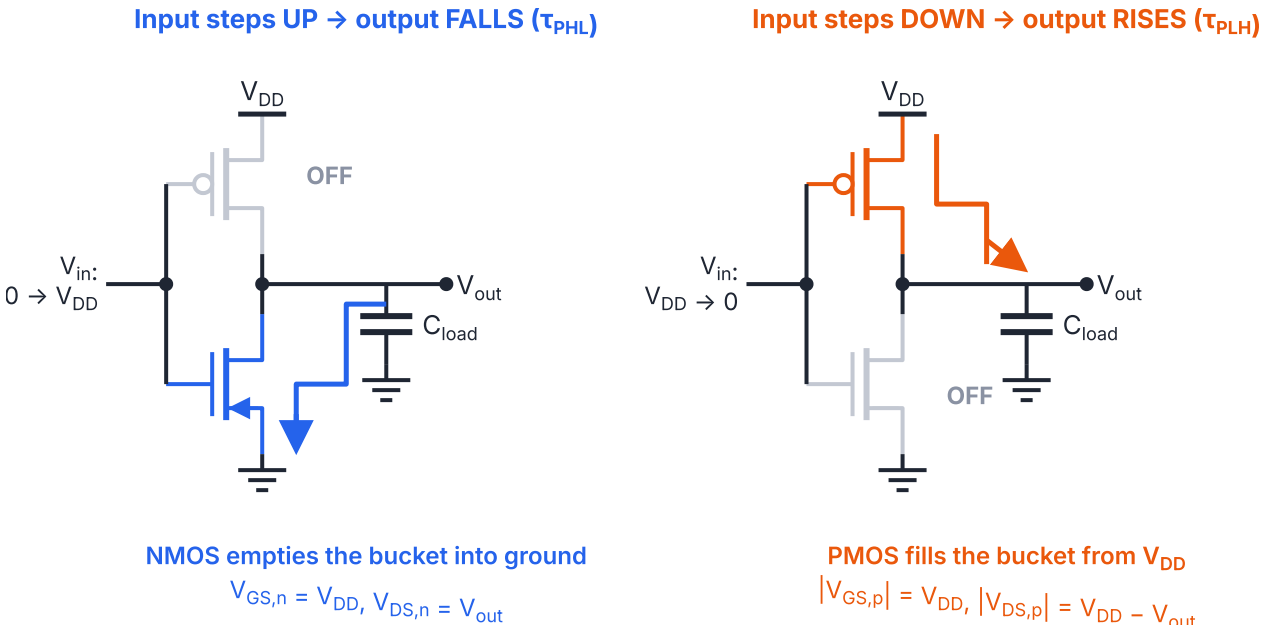


Fig. B2.1 Left: the falling output is set by the NMOS alone. Right: the rising output is set by the PMOS alone. The gate voltage of the working transistor is the full V_{DD} the whole time. Only V_{DS} changes, because V_{DS} is tied to V_{out}.

The key question: which region is the working transistor in?

Take the falling output. The NMOS has $V_{GS} = V_{DD}$, so its overdrive is $u = V_{DD} - V_{T,n}$, a constant. Its $V_{DS} = V_{out}$, which falls. The saturation test is $V_{DS} \geq u$:

- **At the start** $V_{out} = V_{DD}$, which is bigger than $V_{DD} - V_T$ → **saturated**. Current is constant $\frac{k}{2}u^2$.
- **When** V_{out} **drops below** $V_{DD} - V_T$ → **linear**. Current gets smaller as V_{out} falls.
- **At the 50% point** $V_{out} = V_{DD}/2$. This is below $V_{DD} - V_T$ whenever $V_T < V_{DD}/2$ (always, in normal inverters) → **linear**.

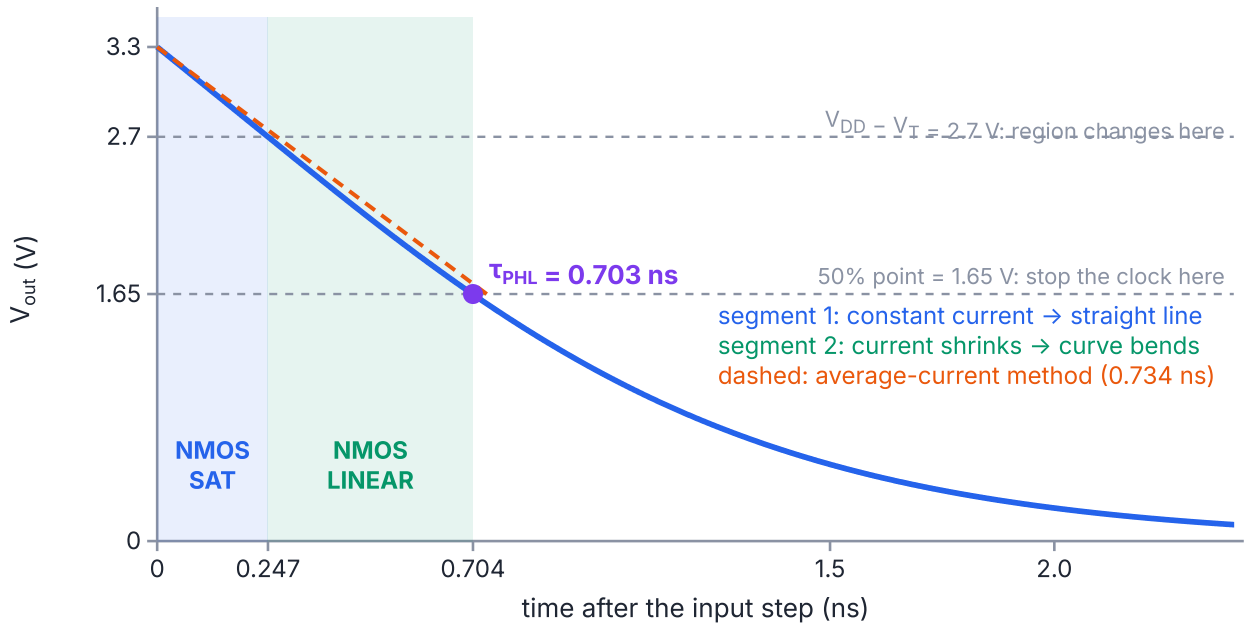


Fig. B2.2 The real fall of Worked Example 25.2 ($V_{DD} = 3.3$ V, $V_{T,n} = 0.6$ V, $k_n = 200$ µA/V², $C = 300$ fF), computed by solving the equation above. First the output drops by exactly V_T in a straight line (saturated, constant current). Then it curves (linear). We stop the clock at the 50% point.

INTUITION

The rise is the same story with the PMOS: $|V_{DS,p}| = V_{DD} - V_{out}$. It starts saturated (output near 0), and goes linear once the output has risen by $|V_{T,p}|$. So **both edges = "saturated for the first V_T volts, then linear"**.

B3 · The definitions

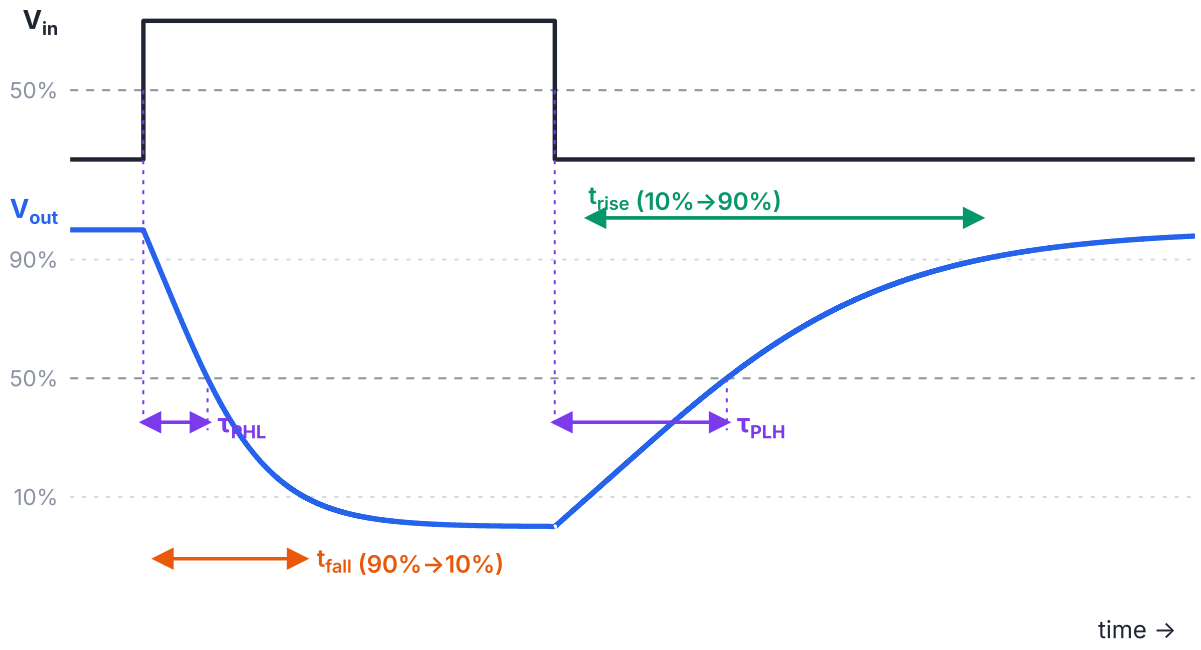


Fig. B3.1 Delays (purple) are measured between **two signals**, input and output, at their 50% points. Rise and fall times (green, orange) are measured on **one signal**, the output, between 10% and 90%.

SYMBOL	MEANING	FORMULA
$V_{50\%}$	halfway voltage	$\frac{V_{OH}+V_{OL}}{2} = \frac{V_{DD}}{2}$ for CMOS
τ_{PHL}	delay, output going **H**igh → **L**ow (NMOS does the work)	$t_{out,50\%} - t_{in,50\%}$
τ_{PLH}	delay, output going **L**ow → **H**igh (PMOS does the work)	same, on the other edge
τ_P	average propagation delay	$\frac{1}{2}(\tau_{PHL} + \tau_{PLH})$
t_{fall}, t_{rise}	how steep the output edge is	time between 90% and 10% of the output swing

MEMORY HOOK

Read τ_{PHL} as "**P**ropagation, output **H**igh to **L**ow". The letters always describe the **output**. Output falling → NMOS → use k_n, $V_{T,n}$. Output rising → PMOS → use k_p, $|V_{T,p}|$.

B4 · Method 1: the average-current method

The idea: the current changes during the fall, but not by much. So pretend it is constant at the average of its value at the start and its value at the 50% point. Then the bucket formula works directly:

$$\tau_{PHL} = \frac{C_{load}\,(V_{DD}/2)}{I_{avg}}, \qquad I_{avg} = \frac{I(\text{start}) + I(50\%\ \text{point})}{2}$$

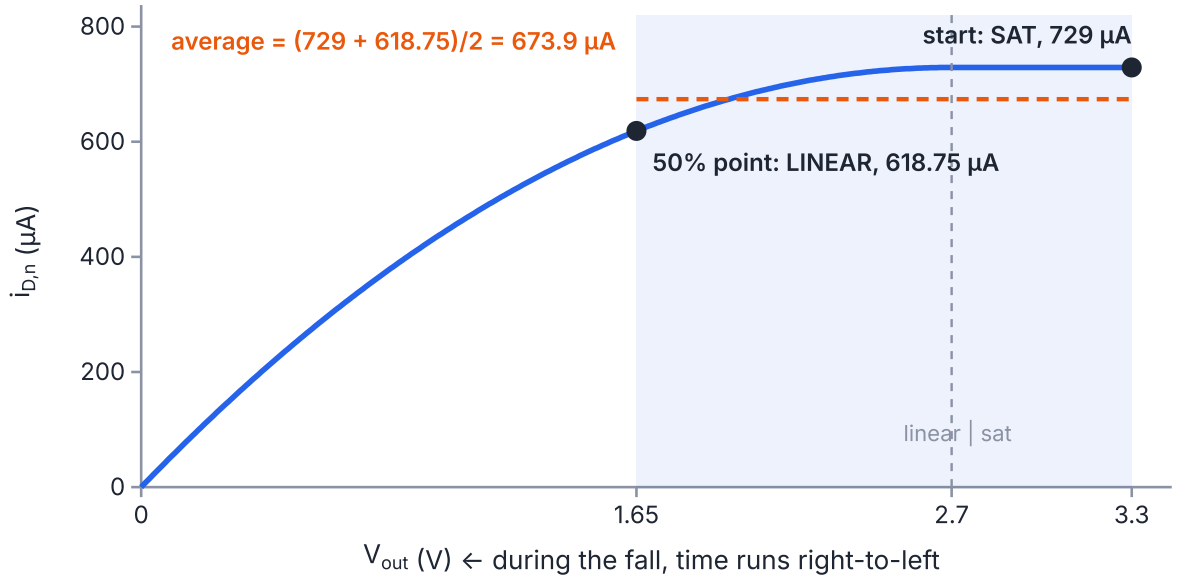


Fig. B4.1 The NMOS current during the fall (Example 25.2). The output starts at the right (3.3 V) and moves left. The method replaces the blue curve between 3.3 V and 1.65 V by the orange line at the average of the two black dots.

THE RECIPE: 4 STEPS, EVERY TIME

1. **Overdrive** of the working transistor: $u = V_{DD} - |V_T|$.
2. **Start point** ($|V_{DS}| = V_{DD}$, saturated): $I_1 = \frac{k}{2}u^2$.
3. **50% point** ($|V_{DS}| = V_{DD}/2$, linear): $I_2 = \frac{k}{2}\left[2u\,\frac{V_{DD}}{2} - \left(\frac{V_{DD}}{2}\right)^2\right]$.
4. $I_{avg} = (I_1 + I_2)/2$, then $\tau = C\,(V_{DD}/2)/I_{avg}$.

MISTAKE NUMBER 1 IN THIS LECTURE

Using the saturation formula at **both** ends. At the 50% point the transistor is **linear**. Check: is $V_{DD}/2 < V_{DD} - |V_T|$? Yes whenever $|V_T| < V_{DD}/2$.

This method is always a few percent **too slow** (≈ 4% in Example 25.2), because the real current curve bulges above the straight average. Use it when the question says "average current", or as a quick check.

B5 · Method 2: the exact formula

Here we do the integral properly. You need the **result** in the exam. Read the derivation once so the formula makes sense.

Step 1: separate the variables

For the fall, only the NMOS works: $C\,\dfrac{dV}{dt} = -I(V)$, so $dt = -\dfrac{C\,dV}{I(V)}$. Time = add up (integrate) $C\,dV/I$ over the voltage range. That is the bucket formula, done in thin slices.

Step 2: segment 1, saturated, from V_{DD} down to $V_{DD} - V_T$

The current is constant, so no integral is needed. The output drops by exactly V_T:

$$t_{sat} = \frac{C \cdot V_T}{\frac{k}{2}u^2} = \frac{2C\,V_T}{k\,u^2}$$

Step 3: segment 2, linear, from u down to $V_{DD}/2$

Now $I = \frac{k}{2}V(2u - V)$. One standard integral (partial fractions):

$$\int \frac{dV}{V(2u - V)} = \frac{1}{2u}\ln\frac{V}{2u - V}$$

Put in the limits $V = u$ (start, where the log is $\ln 1 = 0$) and $V = V_{DD}/2$ (end):

$$t_{lin} = \frac{C}{k\,u}\ln\left(\frac{2u}{V_{DD}/2} - 1\right) = \frac{C}{k\,u}\ln\left(\frac{4u}{V_{DD}} - 1\right)$$

Step 4: add them

MEMORISE: EXACT DELAY OF A CMOS INVERTER (STEP INPUT)

$$\boxed{\tau = \frac{C_{load}}{k\,u}\left[\frac{2|V_T|}{u} + \ln\left(\frac{4u}{V_{DD}} - 1\right)\right], \qquad u = V_{DD} - |V_T|}$$

τ_{PHL}: use k_n and $V_{T,n}$. τ_{PLH}: use k_p and $|V_{T,p}|$.

SAY IT OUT LOUD

"**C over k-u, times [sat bit + log bit].**" Sat bit = $2V_T/u$ (the straight-line part). Log bit = $\ln(4u/V_{DD} - 1)$ (the curved part). The front factor $C/(ku)$ is an RC time constant, because a linear MOSFET near $V_{DS} = 0$ is a resistor $R_{on} = 1/(ku)$.

YOUR LECTURE WRITES IT WITH V_OH AND V_OL

$$\tau_{PHL} = \frac{C_{load}}{k_n(V_{OH} - V_{T0,n})}\left[\frac{2V_{T0,n}}{V_{OH} - V_{T0,n}} + \ln\left(\frac{4(V_{OH} - V_{T0,n})}{V_{OH} + V_{OL}} - 1\right)\right]$$

For CMOS, $V_{OH} = V_{DD}$ and $V_{OL} = 0$, and it becomes the boxed formula. Same thing.

RARE CASE: IF |V_T| ≥ V_DD/2

Then the transistor is still saturated at the 50% point, so there is no linear segment. The whole half-swing is constant current: $\tau = \frac{C\,(V_{DD}/2)}{\frac{k}{2}u^2} = \frac{C\,V_{DD}}{k\,u^2}$. (This is the second line of your formula sheet's $D(k,T)$.)

B6 · Rise and fall time (10% ↔ 90%)

Same integral, different limits: the fall time runs from $0.9V_{DD}$ down to $0.1V_{DD}$, not from V_{DD} to $V_{DD}/2$.

- Saturated piece: from $0.9V_{DD}$ down to u. Only exists if $0.9V_{DD} > u$, i.e. $|V_T| > 0.1V_{DD}$ (normal case). It lasts $\dfrac{C(0.9V_{DD}-u)}{\frac{k}{2}u^2} = \dfrac{C}{ku}\cdot\dfrac{2(|V_T|-0.1V_{DD})}{u}$.
- Linear piece: from u down to $0.1V_{DD}$: $\dfrac{C}{ku}\ln\left(\dfrac{2u}{0.1V_{DD}}-1\right) = \dfrac{C}{ku}\ln\left(\dfrac{20u}{V_{DD}}-1\right)$.

FALL TIME (AND RISE TIME WITH THE PMOS VALUES)

$$t_{fall} = \frac{C}{k\,u}\left[\frac{2(|V_T|-0.1V_{DD})}{u} + \ln\left(\frac{20u}{V_{DD}}-1\right)\right]$$

Same shape as the delay formula. Only the numbers inside change: $V_T \to V_T - 0.1V_{DD}$ and $4 \to 20$.

B7 · One MOSFET + one capacitor (Razavi-type questions)

Sometimes there is no inverter: just one NMOS charging or discharging a capacitor from some starting voltage. The method is the same as B5, but you **must decide the region yourself**, and the limits are whatever the question gives.

GENERAL RECIPE

1. Write V_{GS} and V_{DS} in terms of the capacitor voltage V_X. (Which terminal is the capacitor on?)
2. Find the overdrive $u = V_{GS} - V_T$ and check the region: saturated if $V_{DS} \geq u$.
3. Write $C\,dV_X/dt = \pm I$ and separate: $dt = C\,dV_X/I$.
4. Integrate between the start and end voltages. If the region changes on the way, split at the boundary.

SITUATION	REGION	TIME FROM V_I TO V_F
Cap on drain, gate fixed, $V_{DS} = V_X \geq u$	saturated	$\dfrac{C\,(V_i - V_f)}{\frac{k}{2}u^2}$ (straight line)
Cap on drain, gate fixed, $V_X < u$ the whole time	linear	$\dfrac{C}{k\,u}\ln\dfrac{V_i\,(2u - V_f)}{V_f\,(2u - V_i)}$
Cap on source (follower), $V_{GS} = V_G - V_X$	saturated always (drain at V_{DD})	$\dfrac{2C}{k}\left[\dfrac{1}{V_G - V_T - V_f} - \dfrac{1}{V_G - V_T - V_i}\right]$

The follower slows down as it gets close to $V_G - V_T$, because its overdrive is shrinking to zero. It never quite gets there.

B8 · Design: what makes the gate faster

Every delay formula has the shape $\tau \propto \dfrac{C_{load}}{k\,(V_{DD} - |V_T|)}$. So there are three knobs:

KNOB	EFFECT	CATCH
smaller C_{load}	τ goes down in proportion	shorter wires, fewer gates driven
bigger k (wider W/L)	more current, τ ∝ 1/k	a wider transistor has a bigger drain C_{db}, and a bigger gate that slows the previous stage
bigger V_{DD}	more current	power $\propto V_{DD}^2$ goes up

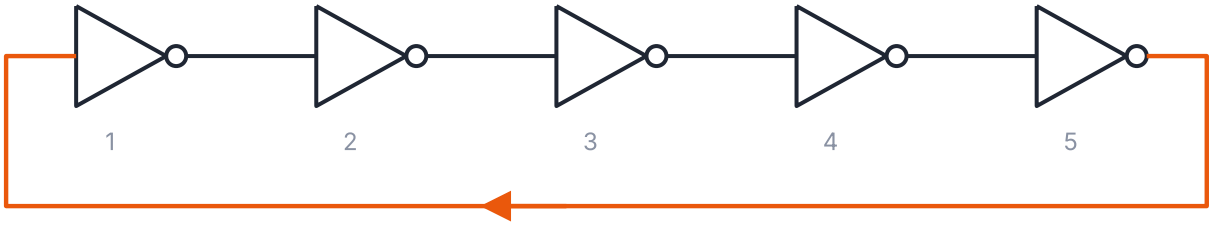
TWO DESIGN RULES YOU WILL USE

Rule 1 (scaling): with C and voltages fixed, $\tau \propto 1/k$. To change a delay by a factor, change k (so W/L) by the inverse factor.

Rule 2 (equal edges): $\tau_{PHL} = \tau_{PLH}$ needs $k_p = k_n$ (if $|V_{T,p}| = V_{T,n}$). Since $\mu_n \approx 2.5\mu_p$, that means $(W/L)_p \approx 2.5\,(W/L)_n$. That is **exactly the symmetric inverter of Lecture 24** ($V_{th} = V_{DD}/2$, equal noise margins). So "sized for equal rise and fall delays" means "symmetric".

B9 · Ring oscillator and dynamic power

Ring oscillator



odd N (here 5): no stable state, so an edge runs round forever

one period = edge goes round TWICE (once rising, once falling) → $T = 2N\tau_p$

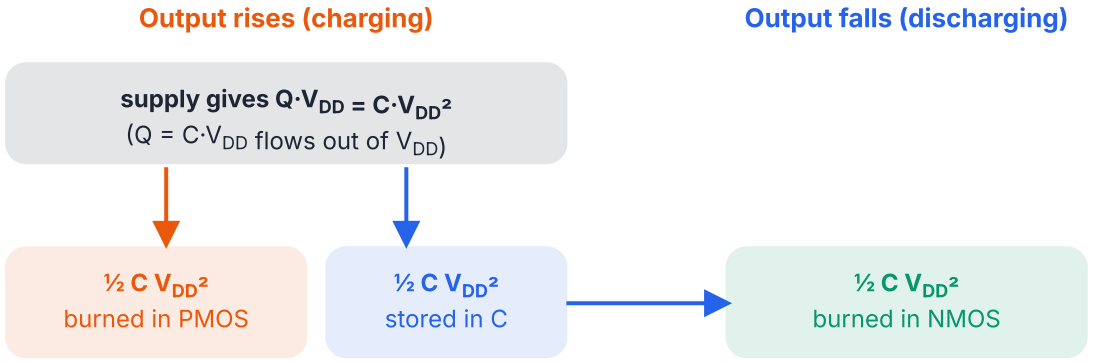
Fig. B9.1 An odd number of inverters in a loop. Each stage adds one delay. To return to the starting state, the edge must go round once as a rising edge and once as a falling edge.

RING OSCILLATOR

$$T = 2N\tau_P, \qquad f = \frac{1}{2N\tau_P}, \qquad N \text{ odd}$$

Used backwards to **measure** τ_P: measure f, then $\tau_P = \dfrac{1}{2Nf}$.

Dynamic power



Per full cycle (up and down): energy $C \cdot V_{DD}^2$, power $P = C \cdot V_{DD}^2 \cdot f$

Fig. B9.2 Charging C to V_{DD} takes CV_{DD}^2 from the supply. Half is stored, half is burned as heat in the PMOS, whatever its size. Discharging burns the stored half in the NMOS.

MEMORISE

$$E_{cycle} = C_{load}V_{DD}^2, \qquad P_{dyn} = C_{load}V_{DD}^2 f, \qquad E_{PMOS} = E_{NMOS} = \tfrac{1}{2}C_{load}V_{DD}^2$$

Static power ≈ 0 in CMOS, because one transistor is always off when the output is steady.

UNIT SHORTCUT THAT SAVES CALCULATOR ERRORS

fF × V ÷ µA = ns and **fF × V ÷ mA = ps**. Put C in fF, k in µA/V², I in µA and the delay comes out in ns directly.

Example: $\dfrac{300\ \text{fF}}{200\,\mu\text{A/V}^2 \times 2.7\ \text{V}} = 0.556$ ns.

Calculator: use **ln**, not log. Type the bracket in one line, e.g. `1.2÷2.7+ln(10.8÷3.3−1)` → 1.2654.

B10 · What Lecture 26 adds

Most of Lecture 26 repeats the Lecture 25 derivation, this time written out in full for τ_{PLH} as well. Three things are new or stated more clearly.

1 · The PLH formula, written out

$$\tau_{PLH} = \frac{C_{load}}{k_p(V_{DD} - |V_{T0,p}|)} \left[\frac{2|V_{T0,p}|}{V_{DD} - |V_{T0,p}|} + \ln\left(\frac{4(V_{DD} - |V_{T0,p}|)}{V_{DD}} - 1 \right) \right]$$

This is the boxed formula of B5 with the PMOS numbers. The PMOS stays saturated while $V_{out} < |V_{T0,p}|$, then goes linear. It is the mirror image of the NMOS story.

2 · Equal delays

CONDITION FOR $T_{PHL} = T_{PLH}$

$$k_n = k_p \quad \text{and} \quad V_{T0,n} = |V_{T0,p}|$$

Same k and same threshold give the same formula, so the same delay. This is the symmetric inverter.

3 · A real input is not a perfect step



A slow input turns the transistor on gradually, so the output starts later → the delay grows.

Fig. B10.1 The lecture formulas assume the input jumps instantly. A real input takes some time τ_r (rise) or τ_f (fall) to change.

DELAY WITH A SLOW INPUT (LECTURE 26)

$$\tau_{PHL}(\text{actual}) = \sqrt{\tau_{PHL}^2(\text{step input}) + \left(\frac{\tau_r}{2}\right)^2} \qquad \tau_{PLH}(\text{actual}) = \sqrt{\tau_{PLH}^2(\text{step input}) + \left(\frac{\tau_f}{2}\right)^2}$$

τ_r is the **input's** rise time (the input rising makes the output fall: PHL). τ_f is the **input's** fall time.

Example: the inverter of Q3 ($\tau_{PHL} = 0.703$ ns) gets an input with $\tau_r = 0.5$ ns: $\sqrt{0.703^2 + 0.25^2} = \sqrt{0.4942 + 0.0625} = 0.746$ ns. That is a little slower than with a perfect step.

4 · Ring oscillator

The lecture ends with three inverters in a loop, each driving a capacitor (C_1, C_2, C_3). That is the ring oscillator of B9: $f = 1/(2N\tau_P)$ with $N = 3$.

R · Lectures 23–24 refresher (what Tutorial 8 needs)

Tutorial 8 is about the **DC** behaviour (Lectures 23–24), not delay. Here is the minimum you need, in the same notation as your guide.

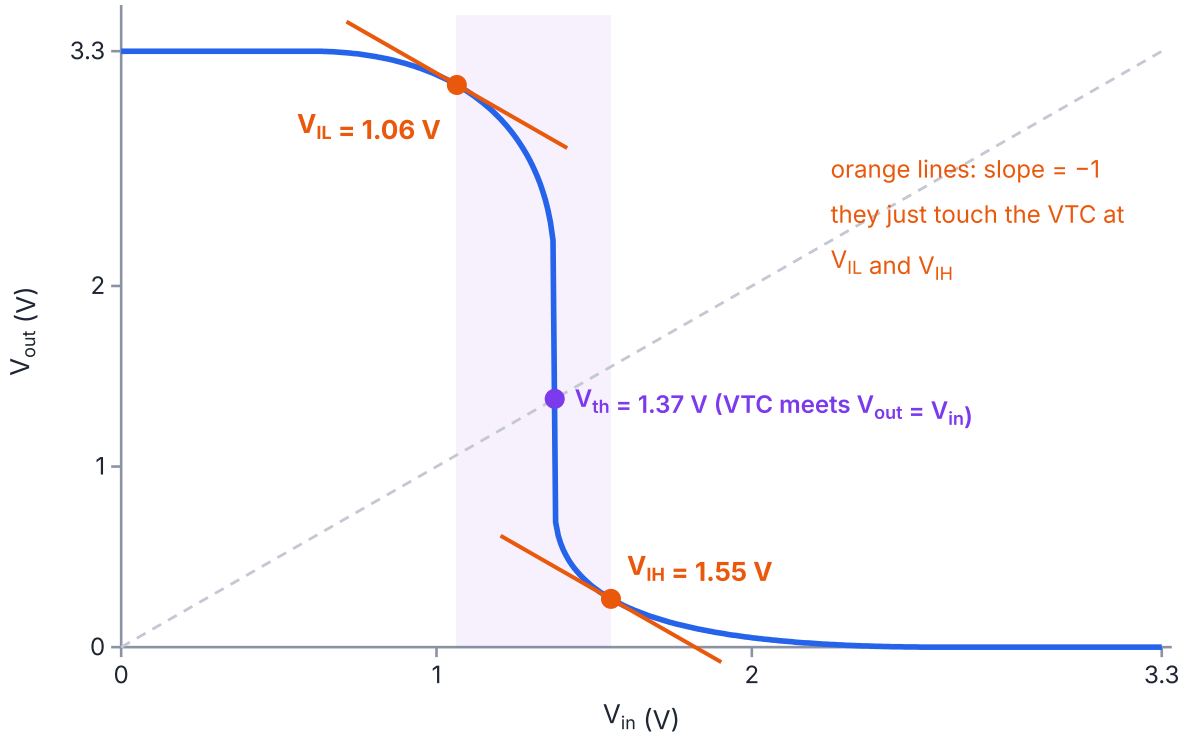


Fig. R.1 The VTC of the Tutorial 8 Q1 inverter (computed). V_{IL} and V_{IH} are the two points where the slope is −1. V_{th} is where the curve crosses the line $V_{out} = V_{in}$.

POINT	WHICH TRANSISTOR IS IN WHICH REGION	EQUATIONS TO USE (FORMULA SHEET)
V_{IL} (input low, output still high)	NMOS **saturated** (just on), PMOS **linear**	① current balance + ② $V_{IL} = \dfrac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R}$
V_{th} (middle)	both **saturated**	$V_{th} = \dfrac{V_{T0,n} + \sqrt{1/k_R}\,(V_{DD} + V_{T0,p})}{1 + \sqrt{1/k_R}}$
V_{IH} (input high, output nearly low)	NMOS **linear**, PMOS **saturated**	③ current balance + ④ $V_{IH} = \dfrac{V_{DD} + V_{T0,p} + k_R(2V_{out} + V_{T0,n})}{1 + k_R}$

THE TWO CURRENT BALANCES

① at V_{IL}: $k_R(V_{in} - V_{T0,n})^2 = 2(V_{in} - V_{DD} - V_{T0,p})(V_{out} - V_{DD}) - (V_{out} - V_{DD})^2$
③ at V_{IH}: $k_R\left[2(V_{in} - V_{T0,n})V_{out} - V_{out}^2\right] = (V_{in} - V_{DD} - V_{T0,p})^2$
$k_R = k_n/k_p$. Noise margins: $NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$, with $V_{OH} = V_{DD}$, $V_{OL} = 0$.

HOW TO SOLVE ① + ② (OR ③ + ④) BY HAND

Put ② into ① (replace every V_{in} by its expression in V_{out}). You get a **quadratic in** V_{out}. Solve it, keep the root that makes sense (near V_{DD} for V_{IL}, near 0 for V_{IH}), then put V_{out} back into ② to get V_{IL}. On a Casio, the equation mode solves the quadratic for you.

DESIGNING FOR A TARGET V_{TH} (BACKWARDS)

$$k_R = \left(\frac{V_{DD} + V_{T0,p} - V_{th}}{V_{th} - V_{T0,n}} \right)^2, \qquad \frac{(W/L)_n}{(W/L)_p} = k_R \cdot \frac{\mu_p C_{ox}}{\mu_n C_{ox}}$$

C · Which type of question is it?

Read the question once. Find its row below. The row tells you the theory and the worked question to copy.

Asks V_{IL}, V_{IH}, noise margins, V_{th}, W_n/W_p for a V_{th}?	DC · Lectures 23–24 (refresher R)	T8 Q1–Q4
Asks C_{load}, or gives C_{ox}, C_j, L_D, C_{db} ...?	TYPE 1 · Find C_{load} (B1)	D Q1, Q2
Inverter + k, V_T, V_{DD}, C; asks τ_{PHL}, τ_{PLH}, τ_P?	TYPE 2 · Delay (B4, B5)	T9 Q2 · D Q3
Asks 10%→90% rise or 90%→10% fall time?	TYPE 3 · Rise / fall time (B6)	T9 Q1, Q5b · D Q4
Gives a target delay or V_{th} and asks W?	TYPE 4 · Design / sizing (B8)	T9 Q3, Q4, Q5a · D Q5, Q6
One transistor + one capacitor, odd limits?	TYPE 5 · Lone MOSFET + C (B7)	T9 Q3 · D Q7, Q8
Odd number of inverters in a loop?	TYPE 6 · Ring oscillator (B9)	D Q9
Input has a rise/fall time (not a step)?	Lecture 26 correction (B10)	B10 example
Asks power or energy?	TYPE 7 · Dynamic power (B9)	D Q10

Fig. C.1 The seven question types in Lecture 25. Many exam questions chain two or three of them: find C_{load} → find the delay → size the PMOS.

THE 5-STEP ROUTINE FOR EVERY QUESTION

1. **What it says:** redraw the circuit. Mark which transistor does the work and what changes.
2. **Which theory:** name the section (B1–B9) and its formula.
3. **Plan:** list the numbers you need first (overdrive u, $V_{DD}/2$, C in fF, k in µA/V²).
4. **Solve:** region check first, then plug in.
5. **Check:** right transistor for the edge? linear at 50%? units? Does the answer make sense (PMOS edge slower if $k_p < k_n$)?

Tutorial 8 · solved step by step

Tutorial 8 uses Lectures 23–24 (the refresher R). Each question has the same five steps. Cover the "Solve" part and try it yourself first.

T8 · Q1 · Noise margins of a non-symmetric inverter

$V_{DD} = 3.3$ *V*, $V_{T0,n} = 0.6$ *V*, $V_{T0,p} = -0.7$ *V*, $k_n = 200$, $k_p = 80$ *µA/V²*. *Find the noise margins. ($k_R = 2.5$, not symmetric.)*

IN PLAIN WORDS

"How much noise can sit on a logic 0 or a logic 1 before the next gate gets confused?" We need V_{IL} and V_{IH}, the two slope = −1 points of the VTC (Fig. R.1).

THEORY

Refresher R. At V_{IL} use ① + ②. At V_{IH} use ③ + ④. Not symmetric, so the shortcut $(3V_{DD} + 2V_T)/8$ is **not** allowed.

PLAN

Write ② and ④ with numbers → substitute → solve two quadratics → back-substitute.

SOLVE

V_{IL} (NMOS sat, PMOS linear)

1. ②: $V_{IL} = \dfrac{2V_{out} - 0.7 - 3.3 + 2.5(0.6)}{3.5} = \dfrac{2V_{out} - 2.5}{3.5}$
2. Put it into ①: $2.5(V_{in} - 0.6)^2 = 2(V_{in} - 2.6)(V_{out} - 3.3) - (V_{out} - 3.3)^2$. Expanding gives $V_{out}^2 + 0.0667\,V_{out} - 9.898 = 0$.
3. Positive root: $V_{out} = 3.113$ V (high, as expected).
4. $V_{IL} = (2 \times 3.113 - 2.5)/3.5 =$ **1.064 V**

V_{IH} (NMOS linear, PMOS sat)

1. ④: $V_{IH} = \dfrac{3.3 - 0.7 + 2.5(2V_{out} + 0.6)}{3.5} = \dfrac{4.1 + 5V_{out}}{3.5}$
2. Put it into ③: $2.5[2(V_{in} - 0.6)V_{out} - V_{out}^2] = (V_{in} - 2.6)^2$. This gives $V_{out}^2 + 2.667\,V_{out} - 0.7843 = 0$.
3. Positive root: $V_{out} = 0.2673$ V (low, as expected).
4. $V_{IH} = (4.1 + 5 \times 0.2673)/3.5 =$ **1.553 V**

Noise margins: $NM_L = V_{IL} - 0 =$ **1.06 V** $\quad NM_H = 3.3 - 1.553 =$ **1.75 V**

CHECK

Regions: at V_{IL}, NMOS $V_{DS} = 3.11 \geq 1.064 - 0.6$ ✓ sat; PMOS $|V_{DS}| = 0.19 < 3.3 - 1.064 - 0.7$ ✓ linear. $NM_L < NM_H$ because the strong NMOS ($k_R = 2.5$) pulls the switching point left ✓. I also swept the VTC numerically: slope −1 at 1.064 V and 1.553 V ✓.

T8 · Q2 · Noise margins and V_{th} from μC_{ox} and W/L

nMOS: $V_{T0,n} = 0.6$ *V,* $\mu_n C_{ox} = 60$ *μA/V²,* $(W/L)_n = 8$. *pMOS:* $V_{T0,p} = -0.7$ *V,* $\mu_p C_{ox} = 25$ *μA/V²,* $(W/L)_p = 12$. $V_{DD} = 3.3$ *V. Find the noise margins and* V_{th}.

IN PLAIN WORDS Same job as Q1, but first build k_n and k_p yourself.

THEORY $k = \mu C_{ox}(W/L)$, then refresher R: ①–④ and the V_{th} formula.

PLAN $k_n = 60 \times 8 = 480$, $k_p = 25 \times 12 = 300$ μA/V² → $k_R = 1.6$. Then exactly as Q1.

SOLVE

1. V_{th}: $\sqrt{1/1.6} = 0.7906$. $V_{th} = \dfrac{0.6 + 0.7906(3.3 - 0.7)}{1 + 0.7906} = \dfrac{2.6555}{1.7906} =$ **1.483 V**
2. $V_{IL} = \dfrac{2V_{out} - 4 + 0.96}{2.6} = \dfrac{2V_{out} - 3.04}{2.6}$ → into ①: $V_{out}^2 + 4.067V_{out} - 21.99 = 0$ → $V_{out} = 3.078$ V → $V_{IL} =$ **1.198 V**
3. $V_{IH} = \dfrac{2.6 + 1.6(2V_{out} + 0.6)}{2.6} = \dfrac{3.56 + 3.2V_{out}}{2.6}$ → into ③: $V_{out}^2 + 6.667V_{out} - 1.839 = 0$ → $V_{out} = 0.2653$ V → $V_{IH} =$ **1.696 V**
4. $NM_L =$ **1.20 V**, $NM_H = 3.3 - 1.696 =$ **1.60 V**

CHECK $V_{IL} < V_{th} < V_{IH}$ ✓ (1.198 < 1.483 < 1.696). k_R is closer to 1 than in Q1, so the two margins are closer together ✓.

T8 · Q3 · Design for V_{th} = 1.4 V, then process variation

$V_{T0,n} = 0.6$, $V_{T0,p} = -0.7$ *V,* $\mu_n C_{ox} = 60$, $\mu_p C_{ox} = 25$ *μA/V²,* $V_{DD} = 3.3$ *V,* $L_n = L_p = 0.8$ *μm. (a) Find* W_n/W_p *for* $V_{th} = 1.4$ *V. (b)* $V_{T0,n}$ *varies ±15 %,* $V_{T0,p}$ *varies ±20 %. Find the highest and lowest* V_{th}.

IN PLAIN WORDS (a) We choose the sizes so the inverter flips at 1.4 V. (b) The factory cannot make the thresholds exactly. How far can V_{th} wander?

THEORY (a) The backwards V_{th} formula (refresher R). (b) The forward V_{th} formula with the extreme thresholds. A tug-of-war picture: a stronger (lower-threshold) NMOS wins earlier → lower V_{th}.

SOLVE

1. (a) $k_R = \left(\dfrac{3.3 - 0.7 - 1.4}{1.4 - 0.6}\right)^2 = \left(\dfrac{1.2}{0.8}\right)^2 = 2.25.$
2. Same L, so $k_R = \dfrac{60\,W_n}{25\,W_p}$ → $\dfrac{W_n}{W_p} = 2.25 \times \dfrac{25}{60} =$ **0.9375**
3. (b) $\sqrt{1/k_R} = 2/3$ stays fixed (widths do not change). Ranges: $V_{T0,n} \in [0.51, 0.69]$ V; $V_{T0,p} \in [-0.84, -0.56]$ V.
4. V_{th} goes **up** when $V_{T0,n}$ goes up (weaker NMOS) and when $|V_{T0,p}|$ goes down (stronger PMOS).
5. Max: $\dfrac{0.69 + \frac{2}{3}(3.3 - 0.56)}{5/3} = \dfrac{0.69 + 1.8267}{1.6667} =$ **1.510 V**
6. Min: $\dfrac{0.51 + \frac{2}{3}(3.3 - 0.84)}{5/3} = \dfrac{0.51 + 1.64}{1.6667} =$ **1.290 V**

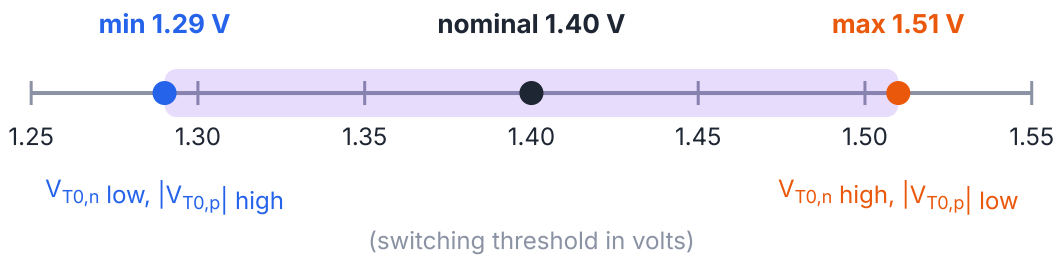
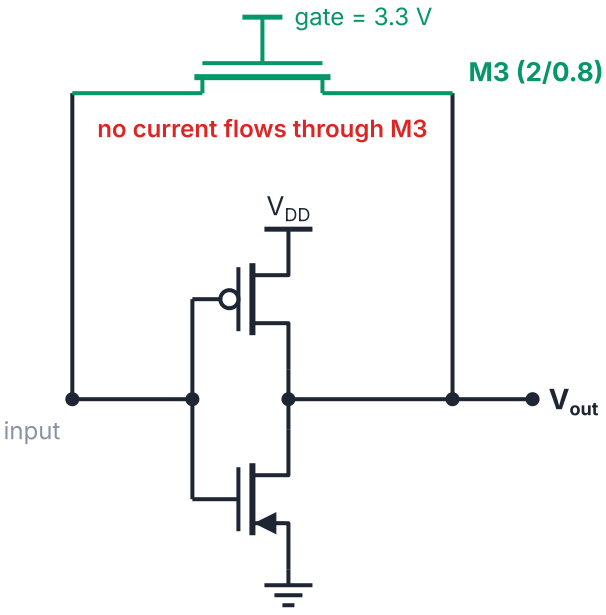


Fig. T8.1 The switching threshold can land anywhere in the purple band, 1.29 V to 1.51 V, about ±0.11 V around the design value.

CHECK Nominal values give back exactly 1.400 V ✓. The two corners are the "both push the same way" combinations ✓.

T8 · Q4 · Inverter with an NMOS feedback switch

The inverter of Q3 has an NMOS M3 (W/L = 2/0.8, gate at 3.3 V) connected from its output back to its input. (a) Find V_{out}. (b) Does the $V_{T0,n}$ variation of M3 change V_{out}? (c) Find the supply current and its variation.



So $V_{in} = V_{out}$: the inverter must sit where its VTC crosses $V_{out} = V_{in}$, which is V_{th}

Fig. T8.2 M3 connects the output back to the input. Its gate is at 3.3 V, so it is a closed switch.

IN PLAIN WORDS We tie the inverter's output to its own input through a switch. The inverter wants to make the output the "opposite" of the input, but they are now the same wire. The only place where "output = input" is allowed is the middle point V_{th}.

THEORY Two facts. ① A MOSFET gate draws **no DC current**, so nothing flows through M3. A transistor that is on but carries no current has $V_{DS} = 0$, so $V_{in} = V_{out}$. ② On the VTC, $V_{out} = V_{in}$ happens only at V_{th}, where both transistors are saturated (Lecture 24).

SOLVE

1. (a) $V_{out} = V_{in} = V_{th} =$ **1.4 V**
2. (b) **No.** M3 carries no current, so it drops no voltage, whatever its threshold. It only has to stay on: $V_{GS,3} = 3.3 - 1.4 = 1.9$ V, which is above even 0.69 V ✓. (The inverter's own threshold variation does move V_{out}, through V_{th}: 1.29 V to 1.51 V as in Q3.)
3. (c) At V_{th} both transistors are saturated and carry the same current from V_{DD} to ground: $I = \frac{\mu_n C_{ox}}{2}\frac{W_n}{L}(V_{th} - V_{T0,n})^2 = \frac{60}{2} \cdot \frac{W_n}{0.8}(0.8)^2 = 24\,W_n$ µA (with W_n in µm).
4. Variation: put each corner of Q3(b) in. Lowest: $V_{T0,n} = 0.69$, $V_{T0,p} = -0.84$ → $V_{th} = 1.398$ V → $18.8\,W_n$ µA. Highest: $V_{T0,n} = 0.51$, $V_{T0,p} = -0.56$ → $V_{th} = 1.402$ V → $29.8\,W_n$ µA. So the current changes by about **−22 % to +24 %**.
5. Example: if $W_p = 2$ µm then $W_n = 1.875$ µm and $I \approx$ **45 µA** (range 35 µA to 56 µA).

NOTE

The tutorial does not give the actual widths, only the ratio from Q3. So the current can only be given per µm of W_n, or for a width you choose. Say this in your answer.

Tutorial 9 · solved step by step

Tutorial 9 is Lectures 25–26: delay, fall and rise time, and design. The theory is in B4–B8.

T9 · Q1 · Fall time by both methods

$V_{DD} = 5$ V, $C_{load} = 1$ pF, $\mu_n C_{ox} = 20$ μA/V², $(W/L)_n = 10$, $V_{Tn} = 1$ V. Find τ_{fall} from $V_{out} = 4.5$ V to 0.5 V by the average-current method and by the differential-equation method.

IN PLAIN WORDS

The input jumps high. The NMOS empties a 1 pF bucket. How long does the water level take to go from 4.5 V down to 0.5 V?

THEORY

B6 (rise/fall time) and B7 (integrate in pieces). The same as the delay, but the limits are 90 % and 10 %, not 100 % and 50 %.

PLAN

$k_n = 20 \times 10 = 200$ μA/V². Overdrive $u = 5 - 1 = 4$ V. The NMOS is saturated while $V_{out} \geq 4$ V, linear below. So 4.5 → 4 V is saturated and 4 → 0.5 V is linear. $C = 1000$ fF.

SOLVE

Average-current method

1. At 4.5 V ($V_{DS} = 4.5 \geq 4$, sat): $I_1 = 100 \times 4^2 = 1600$ μA.
2. At 0.5 V (linear): $I_2 = 100\,[2(4)(0.5) - 0.5^2] = 100 \times 3.75 = 375$ μA.
3. $I_{avg} = (1600 + 375)/2 = 987.5$ μA.
4. $\tau_{fall} = \dfrac{1000\ \text{fF} \times (4.5 - 0.5)\ \text{V}}{987.5\ \mu\text{A}} =$ **4.05 ns**

Differential-equation method

1. Saturated piece, 4.5 → 4 V (constant current 1600 μA): $\dfrac{1000 \times 0.5}{1600} = 0.3125$ ns.
2. Linear piece, 4 → 0.5 V: $\dfrac{C}{ku} \ln \dfrac{V_i(2u - V_f)}{V_f(2u - V_i)} = \dfrac{1000}{200 \times 4} \ln \dfrac{4 \times 7.5}{0.5 \times 4} = 1.25 \times \ln 15 = 1.25 \times 2.708 = 3.385$ ns.
3. $\tau_{fall} = 0.3125 + 3.385 =$ **3.70 ns**

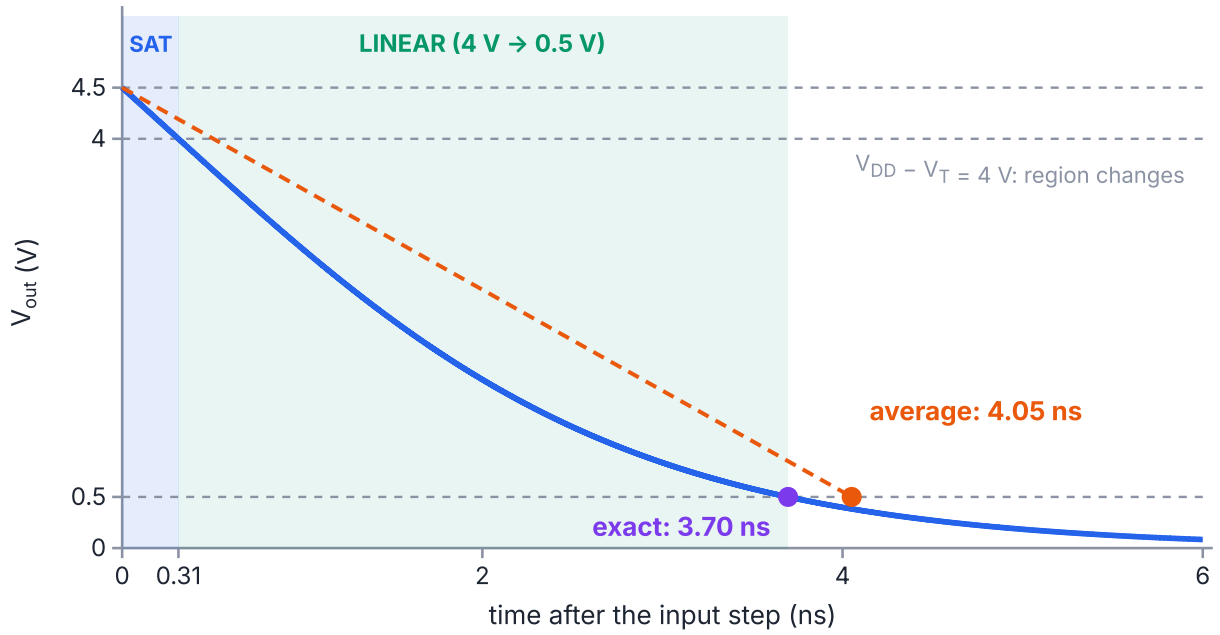


Fig. T9.1 The real fall (blue, computed). It is saturated for only 0.31 ns, then spends a long time in the linear tail. The orange line is the average-current answer.

CHECK

The B6 one-line formula gives the same: $1.25[2(1-0.5)/4 + \ln(20 \times 4/5 - 1)] = 1.25 \times 2.958 = 3.70$ ns ✓. The average method is 10 % too slow here, much worse than for the delay, because the current changes a lot (1600 → 375 µA) over this long swing.

T9 · Q2 · Delay from a measured I–V curve

$V_{DD} = 3.3$ V. With $V_{GS} = 3.3$ V the NMOS reaches $I_{sat} = 2$ mA for $V_{DS} \geq 2.5$ V. Step input 0 → 3.3 V. Find the time for the output to fall from 3.3 V to 1.65 V with $C = 300$ fF.

IN PLAIN WORDS

We are not told k or V_T. We are told what the transistor's current looks like. First work out k and V_T from it, then it is an ordinary τ_{PHL} question.

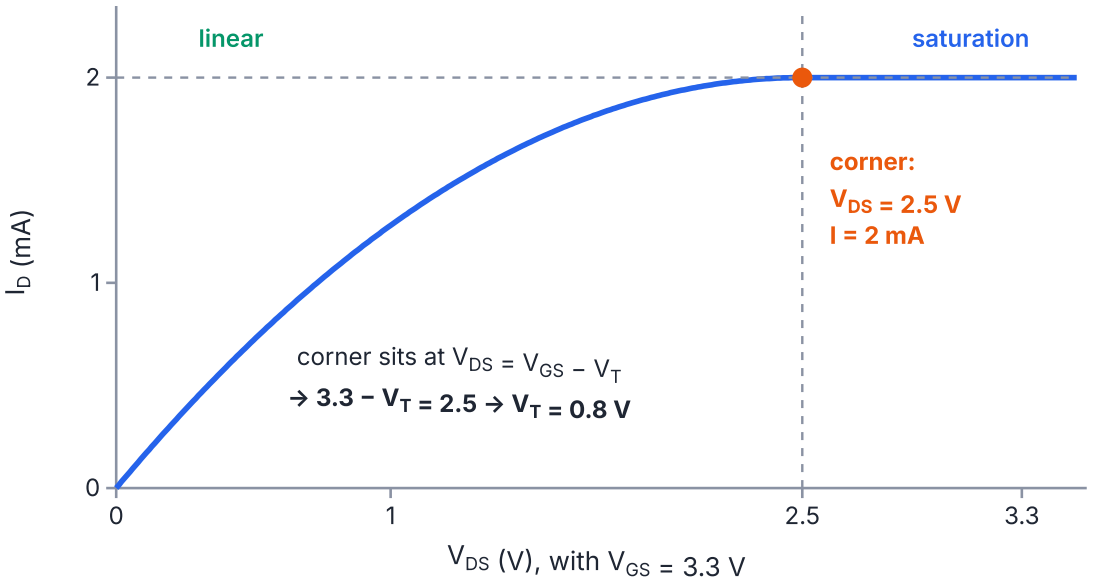


Fig. T9.2 The given data in a picture. The current stops rising (saturates) at the corner $V_{DS} = V_{GS} - V_T$.

THEORY

Square law: saturation starts at $V_{DS} = V_{GS} - V_T$, and $I_{sat} = \frac{k}{2}(V_{GS} - V_T)^2$. Then B5 (exact τ_{PHL}).

SOLVE

1. Corner: $3.3 - V_T = 2.5 \to V_T = 0.8$ V, so $u = 2.5$ V.
2. $2\text{ mA} = \frac{k}{2}(2.5)^2 \to k = \dfrac{4}{6.25} = 0.64$ mA/V² $= 640$ µA/V².
3. Front: $\dfrac{C}{ku} = \dfrac{300}{640 \times 2.5} = 0.1875$ ns.
4. Sat bit $= \dfrac{2(0.8)}{2.5} = 0.64$. Log bit $= \ln\left(\dfrac{4 \times 2.5}{3.3} - 1\right) = \ln 2.0303 = 0.708$.
5. $\tau_{PHL} = 0.1875 \times 1.348 =$ **0.253 ns**

(Average-current check: $I_1 = 2000$ µA, $I_2 = 320[2(2.5)(1.65) - 1.65^2] = 1769$ µA, so $\tau \approx 300 \times 1.65/1884 = 0.263$ ns.)

CHECK

The quick bucket estimate $C\Delta V/I_{sat} = 300 \times 1.65/2000 = 0.248$ ns is close but a bit low, because the current drops in the linear part ✓.

T9 · Q3 · Design an inverter for V_{th} and three timing specs

$\mu_n C_{ox} = 120$, $\mu_p C_{ox} = 60$ µA/V², $L = 0.6$ µm, $V_{T0,n} = 0.8$ V, $V_{T0,p} = -1$ V, $W_{min} = 1.2$ µm, $V_{DD} = 3$ V, $C = 300$ fF. Find W_n, W_p for: $V_{th} = 1.5$ V; $\tau_{PHL} \le 0.2$ ns and $\tau_{PLH} \le 0.15$ ns; a fall from 2 V to 0.5 V in 0.35 ns.

IN PLAIN WORDS

Four rules, two unknowns. The V_{th} rule fixes the **ratio** of the widths. The timing rules each say "the NMOS (or PMOS) must be at least this strong". Find the smallest sizes that obey every rule.

THEORY

Refresher R (backwards $V_{th} \to k_R$). B5 exact delay, solved for k. B7 linear-only integral for the 2 V → 0.5 V fall.

PLAN

① k_R from V_{th}. ② Minimum k_n from each timing rule. ③ The biggest minimum wins. ④ $k \to W$.

SOLVE

1. $k_R = \left(\frac{3-1-1.5}{1.5-0.8}\right)^2 = \left(\frac{0.5}{0.7}\right)^2 = 0.510$. So $k_p = k_n/0.510 = 1.96\,k_n$.
2. $\tau_{PHL} \le 0.2$ ns: $u_n = 2.2$, bracket $= \frac{1.6}{2.2} + \ln(\frac{8.8}{3} - 1) = 0.727 + 0.659 = 1.387$. $k_n \ge \dfrac{300}{0.2 \times 2.2} \times 1.387 = 945$ µA/V².
3. $\tau_{PLH} \le 0.15$ ns: $u_p = 2$, bracket $= \frac{2}{2} + \ln(\frac{8}{3} - 1) = 1 + 0.511 = 1.511$. $k_p \ge \dfrac{300}{0.15 \times 2} \times 1.511 = 1511$ µA/V², so $k_n \ge 1511 \times 0.510 = 771$ µA/V².
4. Fall 2 V → 0.5 V: $2 < u_n = 2.2$, so **linear the whole way**. $t = \dfrac{C}{k_n u} \ln \dfrac{2(4.4-0.5)}{0.5(4.4-2)} = \dfrac{300}{2.2\,k_n} \ln 6.5 \le 0.35$ ns → $k_n \ge 729$ µA/V².
5. Strictest: $k_n \ge 945$ µA/V² (Fig. T9.3). Then $k_p = 945/0.510 = 1853$ µA/V².
6. $W_n = \dfrac{k_n L}{\mu_n C_{ox}} = \dfrac{945 \times 0.6}{120} = 4.73$ µm; $W_p = \dfrac{1853 \times 0.6}{60} = 18.5$ µm. Round up: **$W_n \approx 4.8$ µm, $W_p \approx 18.8$ µm**

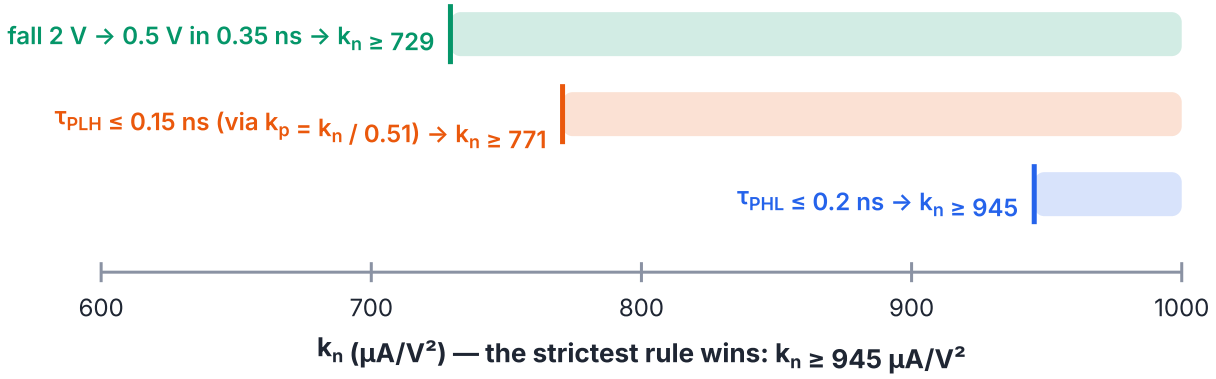


Fig. T9.3 Each spec sets a minimum NMOS strength. Only the strictest one matters.

CHECK

With 4.8 / 18.8 µm: $\tau_{PHL} = 0.197$ ns ✓, $\tau_{PLH} = 0.121$ ns ✓, 2 → 0.5 V in 0.266 ns ✓, $V_{th} = 1.500$ V ✓. Both widths are above $W_{min} = 1.2$ µm ✓.

T9 · Q4 · Design for V_{th} = 2.2 V and τ_{rise} = 5 ns; then lower V_{DD}

nMOS $V_{T0,n} = 0.8$ V, $\mu_n C_{ox} = 50$; pMOS $V_{T0,p} = -1$ V, $\mu_p C_{ox} = 20$ µA/V². $V_{DD} = 5$ V, $L = 1$ µm, $C_{out} = 2$ pF. (a) W_n, W_p for $V_{th} = 2.2$ V and $\tau_{rise} = 5$ ns. (b) τ_P. (c) What changes at $V_{DD} = 3.3$ V?

IN PLAIN WORDS

The rise time is set by the PMOS alone, so it fixes W_p. The V_{th} rule then fixes W_n through the ratio.

THEORY

B6 rise-time formula (10 % → 90 %, PMOS numbers). Refresher R for k_R. B5 for the delays.

SOLVE

1. PMOS: $u = 5 - 1 = 4$ V, $0.1V_{DD} = 0.5$ V. Bracket $= \dfrac{2(1-0.5)}{4} + \ln\left(\dfrac{20\times 4}{5} - 1\right) = 0.25 + \ln 15 = 2.958$.
2. $\tau_{rise} = \dfrac{2000}{k_p \times 4} \times 2.958 = 5$ ns → $k_p = 295.8$ µA/V² → $W_p = 295.8/20 =$ **14.8 µm**
3. $k_R = \left(\dfrac{5-1-2.2}{2.2-0.8}\right)^2 = \left(\dfrac{1.8}{1.4}\right)^2 = 1.653$ → $k_n = 1.653 \times 295.8 = 489$ µA/V² → $W_n = 489/50 =$ **9.8 µm**
4. (b) $\tau_{PHL} = \dfrac{2000}{489 \times 4.2}\left[\dfrac{1.6}{4.2} + \ln(3.36 - 1)\right] = 0.974 \times 1.240 = 1.21$ ns. $\tau_{PLH} = \dfrac{2000}{295.8 \times 4}\left[0.5 + \ln 2.2\right] = 1.690 \times 1.289 = 2.18$ ns. $\tau_P =$ **1.69 ns**
5. (c) Same sizes, $V_{DD} = 3.3$ V. $V_{th} = \dfrac{0.8 + 0.778(3.3-1)}{1.778} =$ **1.46 V**. $\tau_{PHL} = 2.21$ ns, $\tau_{PLH} = 4.26$ ns, $\tau_P =$ **3.23 ns** (about 1.9× slower).

INTERPRETATION (C)

V_{th} scales down with the supply: $2.2/5 = 0.44$ and $1.46/3.3 = 0.44$, so the inverter still switches at about the same *fraction* of V_{DD}. The delays almost double: the charge to move (CV_{DD}) shrinks only in proportion to V_{DD}, but the current shrinks like $(V_{DD} - V_T)^2$. Less supply means a weaker tap. (If your class uses the average-current method for τ_{rise}, you get $k_p = 324$ µA/V², $W_p = 16.2$ µm, $W_n = 10.7$ µm. Say which method you used.)

T9 · Q5 · Sizing when the transistors add to their own load

$V_{th} = 2.4$ V, $V_{DD} = 5$ V, nMOS 0.8 V and 50 µA/V², pMOS −1 V and 20 µA/V², $L = 1$ µm. $C_{out} = 500$ fF + $C_{db,n} + C_{db,p}$, with $C_{db,n} = 100$ fF + $9W_n$ and $C_{db,p} = 80$ fF + $7W_p$ (W in µm). (a) Find the widths so that $\tau_{PHL} < 0.825$ ns. (b) With $(W/L)_n = 6$, $(W/L)_p = 15$ and $C = 250$ fF, find the rise and fall times by the average-current method.

IN PLAIN WORDS

Making the NMOS wider gives more current, but it also makes the bucket bigger (its drain capacitance grows with W). So we must find the width where these two effects balance at 0.825 ns.

THEORY

Refresher R (k_R from V_{th}), B1 (C_{load} adds up), B5 exact delay. (b) is B4 with the 10 % / 90 % points.

SOLVE

1. $k_R = \left(\dfrac{5-1-2.4}{2.4-0.8}\right)^2 = \left(\dfrac{1.6}{1.6}\right)^2 = 1$ → $k_n = k_p$ → $50W_n = 20W_p$ → $W_p = 2.5\,W_n$.
2. $C_{out} = 500 + 100 + 9W_n + 80 + 7(2.5W_n) = 680 + 26.5\,W_n$ fF.
3. Bracket (NMOS, $u = 4.2$): $\dfrac{1.6}{4.2} + \ln(3.36 - 1) = 0.381 + 0.859 = 1.2396$.
4. $\tau_{PHL} = \dfrac{680 + 26.5W_n}{50W_n \times 4.2} \times 1.2396 < 0.825$ → $843.0 + 32.85W_n < 173.25W_n$ → $W_n > 6.00$ µm.

5. So **$W_n \geq 6$ µm, $W_p \geq 15$ µm** (that is why part (b) uses 6 and 15).
6. (b) $k_n = 50 \times 6 = 300$, $k_p = 20 \times 15 = 300$ µA/V². 10 % = 0.5 V, 90 % = 4.5 V, swing 4 V, $C = 250$ fF.
7. Fall (NMOS, $u = 4.2$): at 4.5 V sat: $150 \times 4.2^2 = 2646$ µA; at 0.5 V linear: $150[2(4.2)(0.5) - 0.25] = 592.5$ µA; average 1619 µA. $\tau_{fall} = \dfrac{250 \times 4}{1619} =$ **0.62 ns**
8. Rise (PMOS, $u = 4$): at $V_{out} = 0.5$ ($|V_{DS}| = 4.5$, sat): $150 \times 16 = 2400$ µA; at $V_{out} = 4.5$ ($|V_{DS}| = 0.5$, linear): $150[2(4)(0.5) - 0.25] = 562.5$ µA; average 1481 µA. $\tau_{rise} = \dfrac{1000}{1481} =$ **0.68 ns**

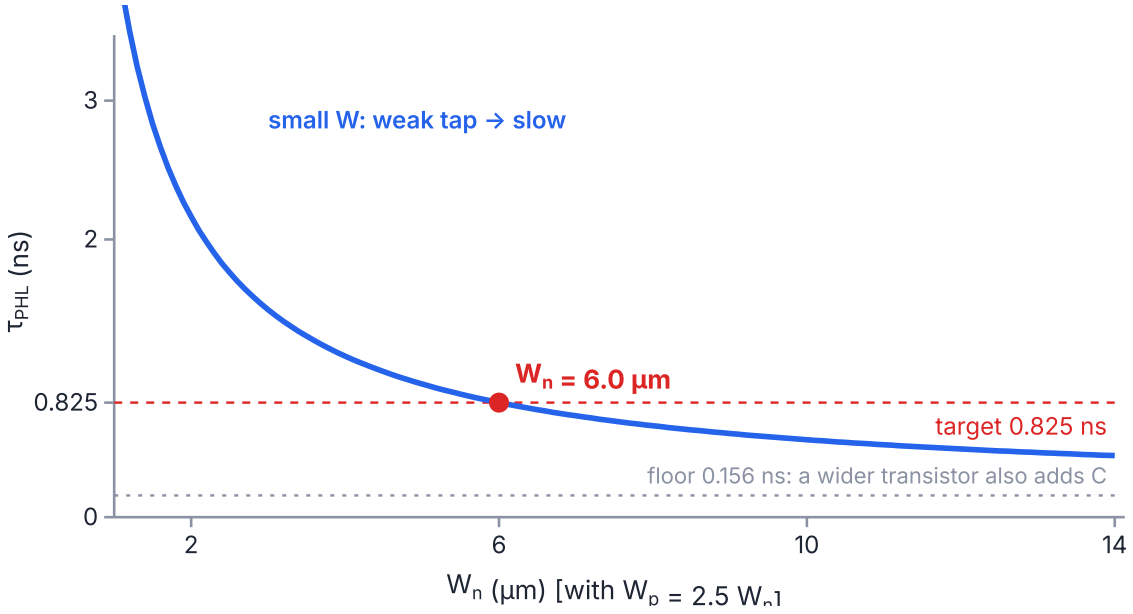


Fig. T9.4 Delay against NMOS width when the transistors load themselves. Wider helps at first, then flattens: the delay can never drop below 0.156 ns, because each extra µm adds both current and capacitance.

CHECK At $W_n = 6$: $C = 839$ fF and $\tau_{PHL} = 0.825$ ns ✓. Rise is slower than fall because the PMOS has the bigger threshold (less overdrive) ✓.

WHAT IN THESE TUTORIALS GOES BEYOND YOUR LECTURE NOTES

1. **10 %–90 % rise and fall times** (T9 Q1, Q4, Q5b). The lectures define them but only derive the 50 % delay. You get them with the same integral and different limits (section B6). This is the most important "extra".
2. **Average-current method for rise/fall time** (T9 Q1, Q5b). Same idea as B4, but average the currents at the 90 % and 10 % points.
3. **A fall between any two voltages** (T9 Q3: 2 V → 0.5 V). Uses the lone-segment integral of B7.
4. **Reading k and V_T from an I–V description** (T9 Q2). No new theory, just the square law read backwards.
5. **T8 Q4** uses the fact that a MOSFET gate draws no DC current, and its part (c) cannot give a number in µA without a width the tutorial does not state.
6. **Lecture 26's slow-input formula** is not needed in either tutorial, but it could appear in the quiz.

D · More practice: ten more worked questions

Q1 · Find C_{load} (Type 1)

$C_{ox} = 3.8$ fF/μm², $L_D = 0.08$ μm, $L = 0.5$ μm, $W_n = 1$ μm, $W_p = 2.5$ μm, $C_{db,n} = 2$ fF, $C_{db,p} = 4$ fF, $C_{int} = 5$ fF. The next stage is identical. Find C_{load}. Which term is biggest?

WHAT IT SAYS

An inverter drives an identical inverter (Fig. B1.1). We must add up every capacitor that touches the output. The junction caps and the wire are given. We must work out the overlaps and the next gate.

THEORY

B1: the plate test. $C_{gd} = C_{ox} W L_D$ for each transistor; $C_g = C_{ox}(W_n L + W_p L)$ for the next stage.

PLAN

Compute 3 missing terms, then add 6 terms.

SOLVE

1. $C_{gd,n} = 3.8 \times 1 \times 0.08 = 0.304$ fF
2. $C_{gd,p} = 3.8 \times 2.5 \times 0.08 = 0.76$ fF
3. $C_g = 3.8 \times (1 + 2.5) \times 0.5 = 6.65$ fF
4. $C_{load} = 0.304 + 0.76 + 2 + 4 + 5 + 6.65 =$ **18.7 fF**

Biggest: the next stage's gate (6.65 fF), then the wire (5 fF).

CHECK

No C_{gs} of this stage added ✓. Units fF/μm² × μm × μm = fF ✓. (Some books count C_{gd} twice for the Miller effect → 19.8 fF. The lecture counts it once.)

Q2 · C_{load} from a process table, then a quick delay (Type 1)

$(W/L)_n = 5/0.5$, $(W/L)_p = 10/0.5$, driving an identical inverter. $C_{ox} = 3.84$ fF/μm². C_{GDO} = 0.4 (n), 0.3 (p) fF/μm. C_j = 0.56 (n), 0.94 (p) fF/μm². C_{jsw} = 0.035 (n), 0.032 (p) fF/μm. Drain length $E = 1.5$ μm, zero bias, no wire. Find C_{load}, then estimate the delay with $T_d \approx CV_{DD}/I$ using the NMOS saturation current ($V_{DD} = 3$ V, $\mu_n C_{ox} = 134.3$ μA/V², $V_{TH} = 0.7$ V).

WHAT IT SAYS

Same as Q1, but nothing is given directly: every term comes from per-μm or per-μm² numbers. Then a rough "bucket" delay.

THEORY

B1: $C_{gd} = C_{GDO} W$; $C_{db} = WEC_j + 2(W + E)C_{jsw}$ (Fig. B1.2); $C_g = C_{ox}[(WL)_n + (WL)_p]$. Then A.3: $\tau \approx C\Delta V/I$.

PLAN

5 terms (no wire), using n values for the NMOS and p values for the PMOS. Then $I = \frac{k}{2}u^2$ with $k = \mu_n C_{ox}(W/L)$.

SOLVE

1. Overlaps: $C_{gd,n} = 0.4 \times 5 = 2.0$ fF; $C_{gd,p} = 0.3 \times 10 = 3.0$ fF.
2. $C_{db,n} = 5(1.5)(0.56) + 2(5 + 1.5)(0.035) = 4.2 + 0.455 = 4.66$ fF.

3. $C_{db,p} = 10(1.5)(0.94) + 2(10 + 1.5)(0.032) = 14.1 + 0.736 = 14.84$ fF.
4. $C_g = 3.84 \times (5 \times 0.5 + 10 \times 0.5) = 3.84 \times 7.5 = 28.8$ fF.
5. $C_{load} = 2 + 3 + 4.66 + 14.84 + 28.8 =$ **53.3 fF**
6. $k_n = 134.3 \times 10 = 1343$ µA/V²; $I = \frac{1343}{2}(3 - 0.7)^2 = 671.5 \times 5.29 = 3552$ µA $= 3.55$ mA.
7. $T_d \approx \dfrac{53.3\text{ fF} \times 3\text{ V}}{3.55\text{ mA}} =$ **≈ 45 ps** (fF·V/mA = ps).

CHECK

$W/L = 5/0.5 = 10$, so multiply $\mu_n C_{ox}$ by 10 ✓. PMOS used 0.94 and 0.032 ✓. The PMOS drain junction is the 2nd-biggest term: wide transistors are not free.

Q3 · Delays by both methods (Type 2) · Worked Example 25.2

$V_{DD} = 3.3$ V, $C_{load} = 0.3$ pF, $k_n = 200$ µA/V², $k_p = 80$ µA/V², $V_{T0,n} = 0.6$ V, $V_{T0,p} = -0.7$ V. Find τ_{PHL}, τ_{PLH} and τ_P by the average-current method and by the exact method.

WHAT IT SAYS

A CMOS inverter drives a 300 fF bucket. Falling edge: NMOS empties it (Fig. B2.1 left). Rising edge: PMOS fills it (right). Find how long each takes to reach 1.65 V.

THEORY

B4 (average current) and B5 (exact formula). Region check from B2: saturated at the start, linear at 50%.

PLAN

Numbers first: $u_n = 3.3 - 0.6 = 2.7$ V; $u_p = 3.3 - 0.7 = 2.6$ V; $V_{DD}/2 = 1.65$ V; $C = 300$ fF. Answers come out in ns.

SOLVE

A · Average current, fall (NMOS)

1. Start, $V_{DS} = 3.3 \geq 2.7$ → sat: $I_1 = 100 \times 2.7^2 = 729$ µA.
2. 50%, $V_{DS} = 1.65 < 2.7$ → linear: $I_2 = 100\,[2(2.7)(1.65) - 1.65^2] = 100 \times 6.1875 = 618.75$ µA.
3. $I_{avg} = 673.9$ µA → $\tau_{PHL} = 300 \times 1.65/673.9 =$ **0.734 ns**

B · Average current, rise (PMOS)

1. Start: $I_1 = 40 \times 2.6^2 = 270.4$ µA. 50%: $I_2 = 40\,[2(2.6)(1.65) - 2.7225] = 234.3$ µA.
2. $I_{avg} = 252.35$ µA → $\tau_{PLH} = 495/252.35 =$ **1.96 ns**

C · Exact, fall

1. Front: $\dfrac{C}{k_n u} = \dfrac{300}{200 \times 2.7} = 0.5556$ ns.
2. Sat bit $= 2(0.6)/2.7 = 0.4444$. Log bit $= \ln(4 \times 2.7/3.3 - 1) = \ln 2.2727 = 0.8210$.
3. $\tau_{PHL} = 0.5556 \times 1.2654 =$ **0.703 ns** (0.247 ns sat + 0.456 ns linear)

D · Exact, rise

1. Front: $\dfrac{300}{80 \times 2.6} = 1.4423$ ns. Sat bit $= 1.4/2.6 = 0.5385$. Log bit $= \ln(10.4/3.3 - 1) = \ln 2.1515 = 0.7662$.

2. $\tau_{PLH} = 1.4423 \times 1.3047 =$ **1.88 ns**

E · Average: $\tau_P = \frac{1}{2}(0.703 + 1.882) =$ **1.29 ns** (exact). By average current: $\frac{1}{2}(0.734 + 1.96) = 1.35$ ns.

CHECK

Average-current is ≈ 4% above exact on both edges ✓ (it always overestimates). The rise is slower because $k_p < k_n$ ✓. I checked 0.703 and 1.882 ns by solving the differential equation numerically.

Q4 · Fall time and rise time (Type 3)

For the inverter of Q3, find the output fall time (90% → 10%) and rise time (10% → 90%).

WHAT IT SAYS

Now we time only the output, from 2.97 V to 0.33 V (fall), and from 0.33 V to 2.97 V (rise). No input involved.

THEORY

B6: same integral as the delay, but new limits. Check $|V_T| > 0.1V_{DD} = 0.33$ V → yes for both, so both edges have a saturated piece.

PLAN

Reuse $C/(ku)$ from Q3. New sat bit $2(V_T - 0.33)/u$. New log bit $\ln(20u/V_{DD} - 1)$.

SOLVE

1. Fall: sat bit $= 2(0.6 - 0.33)/2.7 = 0.200$; log bit $= \ln(20 \times 2.7/3.3 - 1) = \ln 15.364 = 2.732$.
   $t_{fall} = 0.5556 \times 2.932 =$ **1.63 ns**
2. Rise: sat bit $= 2(0.7 - 0.33)/2.6 = 0.2846$; log bit $= \ln(20 \times 2.6/3.3 - 1) = \ln 14.758 = 2.692$.
   $t_{rise} = 1.4423 \times 2.976 =$ **4.29 ns**

CHECK

Fall time > delay (0.703 ns) ✓, because 10% is deep in the slow linear tail. Ratio $t_{rise}/t_{fall} \approx 2.6$, close to τ_{PLH}/τ_{PHL} ✓. Verified by numerical simulation.

Q5 · Design for a target delay (Type 4)

$V_{DD} = 3.3$ *V*, $V_{T0,n} = 0.6$ *V*, $V_{T0,p} = -0.7$ *V*, $\mu_n C_{ox} = 100$ *μA/V²*, $\mu_p C_{ox} = 40$ *μA/V²*, $C_{load} = 200$ *fF*. *Find* $(W/L)_n$ *and* $(W/L)_p$ *so that* $\tau_{PHL} = \tau_{PLH} = 0.5$ *ns (exact method).*

WHAT IT SAYS

The delay is now given and k is the unknown. Work the exact formula backwards.

THEORY

B5 exact formula, rearranged: $k = \dfrac{C}{\tau\, u}\left[\text{sat bit} + \text{log bit}\right]$. Then $W/L = k/(\mu C_{ox})$.

PLAN

The bracket depends only on voltages, so compute it first for each transistor.

SOLVE

1. NMOS: $u = 2.7$, bracket $= 0.4444 + 0.8210 = 1.2654$ (same as Q3).
   $k_n = \dfrac{200}{0.5 \times 2.7} \times 1.2654 = 187.5$ μA/V² → $(W/L)_n = 187.5/100 =$ **1.87**

2. PMOS: $u = 2.6$, bracket $= 1.3047$.
$$k_p = \frac{200}{0.5 \times 2.6} \times 1.3047 = 200.7 \text{ μA/V²} \rightarrow (W/L)_p = 200.7/40 = \boxed{5.02}$$

CHECK

Units: fF ÷ (ns × V) = μA/V² (the bracket has no units) ✓. Plug k_n back into Q3's formula: $\frac{200}{187.5 \times 2.7} \times 1.2654 = 0.500$ ns ✓. Ratio $5.02/1.87 = 2.68$, a bit more than $\mu_n/\mu_p = 2.5$ because the PMOS has the bigger threshold ✓.

Q6 · Equal rise and fall (Type 4)

For the inverter of Q3, by what factor must $(W/L)_p$ change so that $\tau_{PLH} = \tau_{PHL}$? Assume C_{load} does not change.

WHAT IT SAYS

The PMOS edge (1.882 ns) is too slow. Make the PMOS wider until it matches the NMOS edge (0.703 ns).

THEORY

B8 rule 1: with C and voltages fixed, $\tau_{PLH} \propto 1/k_p$.

PLAN

Factor = old delay ÷ target delay.

SOLVE

Factor $= 1.882/0.703 = \boxed{2.68}$ → $k_p = 80 \times 2.68 = 214$ μA/V², so $(W/L)_p$ is 2.68× bigger.

CHECK (TRAP)

Why not just $k_p = k_n$ (2.5×)? With $k_p = 200$: $\tau_{PLH} = \frac{300}{200 \times 2.6} \times 1.3047 = 0.753$ ns, still slower than 0.703. The PMOS has the bigger threshold, so it needs a bit more width. If the thresholds were equal, $k_p = k_n$ would be exact.

Q7 · An NMOS discharging a capacitor (Type 5) · Razavi 2.11(b)

NMOS: $W/L = 50/0.5$, $\mu_n C_{ox} = 134.3$ μA/V², $V_{TH} = 0.7$ V, $\lambda = \gamma = 0$. Source grounded, drain is node X. $C_1 = 1$ pF from X to ground, starting at 1 V. At $t = 0$ the gate steps from 0 to 3 V. Sketch $V_X(t)$ and find when $V_X = 0.5$ V.

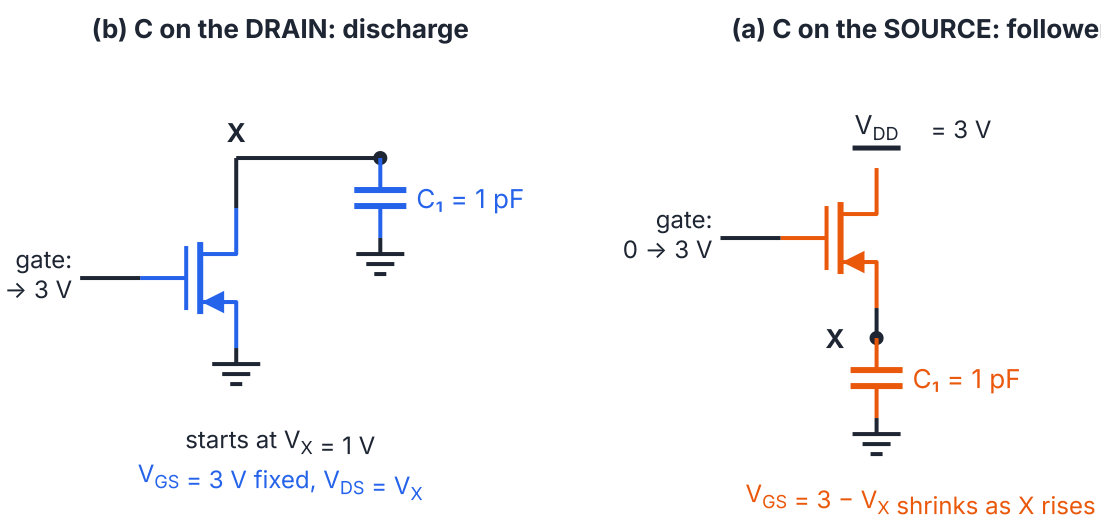


Fig. Q7.1 (b) is this question. (a) is Q8. The only difference: is the capacitor on the drain or on the source?

WHAT IT SAYS

The cap is on the drain. The gate is fixed at 3 V, so $V_{GS} = 3$ V does not change. Only $V_{DS} = V_X$ changes, from 1 V to 0.5 V.

THEORY

B7. Region check: $u = 3 - 0.7 = 2.3$ V. $V_{DS} \le 1 < 2.3$ → **linear the whole time**. So it is "segment 2 only", with a start of 1 V.

PLAN

$t = \dfrac{C}{ku} \ln \dfrac{V_i(2u - V_f)}{V_f(2u - V_i)}$ with $V_i = 1$, $V_f = 0.5$. First find k.

SOLVE

1. $k = 134.3 \times \frac{50}{0.5} = 134.3 \times 100 = 13\,430$ µA/V² $= 13.43$ mA/V².
2. $\dfrac{C}{ku} = \dfrac{1\text{ pF}}{13.43 \times 2.3\text{ mA/V}} = 32.4$ ps.
3. $\ln \dfrac{1 \times (4.6 - 0.5)}{0.5 \times (4.6 - 1)} = \ln \dfrac{4.1}{1.8} = \ln 2.278 = 0.823.$
4. $t = 32.4 \times 0.823 =$ **26.6 ps**

Sketch: starts at 1 V, falls toward 0 like an exponential with time constant $R_{on}C = 32.4$ ps (blue curve below).

CHECK (TRAP)

$W/L = 100$, not 50 ✓. Close to a pure RC fall ($32.4 \ln 2 = 22.4$ ps) but a bit slower, because the $-V^2$ term reduces the current ✓.

Q8 · A source follower charging a capacitor (Type 5) · Razavi 2.11(a)

Same NMOS. Now the drain is at $V_{DD} = 3$ V and C_1 (starting at 1 V) is on the source, node X. The gate steps from 0 to 3 V. Sketch $V_X(t)$, give its final value, and find when $V_X = 2$ V.

WHAT IT SAYS

Fig. Q7.1(a). The cap is on the **source**. As X rises, $V_{GS} = 3 - V_X$ gets smaller, so the transistor slowly turns itself off.

THEORY

B7. Region: $V_{DS} = 3 - V_X = V_{GS} \ge V_{GS} - V_{TH}$ always → **saturated throughout**. Current $= \frac{k}{2}(2.3 - V_X)^2$, which reaches 0 at $V_X = 2.3$ V.

PLAN $C\dfrac{dV_X}{dt} = \dfrac{k}{2}(2.3 - V_X)^2$ → separate → integrate $(2.3 - V)^{-2}$, which gives $\dfrac{1}{2.3 - V}$.

SOLVE

1. $t = \dfrac{2C}{k}\left[\dfrac{1}{2.3 - V_X} - \dfrac{1}{2.3 - 1}\right]$.
2. At $V_X = 2$: bracket $= \dfrac{1}{0.3} - \dfrac{1}{1.3} = 3.333 - 0.769 = 2.564$ V^{-1}.
3. $\dfrac{2C}{k} = \dfrac{2 \times 1\ \text{pF}}{13.43\ \text{mA/V}^2} = 148.9$ ps·V.
4. $t = 148.9 \times 2.564 =$ **0.38 ns**. Final value: **2.3 V** ($= V_G - V_{TH}$), approached but never reached.

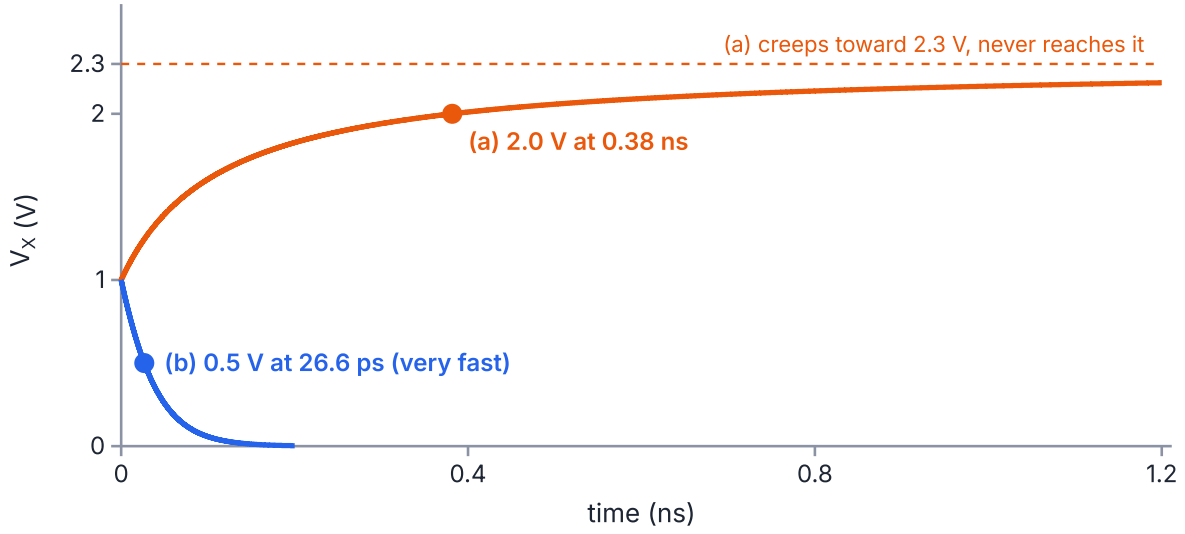


Fig. Q8.1 The two answers on one time axis. The discharge (blue, Q7) is over in tens of ps. The follower (orange, Q8) is about 14× slower and creeps toward 2.3 V.

CHECK The follower slows down near the end because its current falls as a square ✓. With body effect ($\gamma > 0$), V_{TH} rises as X rises, so the final value would be lower than 2.3 V.

Q9 · Ring oscillator (Type 6)

(a) A ring of 5 inverters, each with $\tau_P = 50$ ps. Find the frequency. (b) A 7-stage ring oscillates at 400 MHz. Find τ_P.

WHAT IT SAYS Fig. B9.1. (a) forwards: delay → frequency. (b) backwards: frequency → delay.

THEORY B9: $T = 2N\tau_P$ (the edge goes round twice).

SOLVE

(a) $T = 2 \times 5 \times 50 = 500$ ps → $f = 1/500$ ps $=$ **2 GHz**

(b) $\tau_P = \dfrac{1}{2Nf} = \dfrac{1}{2 \times 7 \times 400 \times 10^6} =$ **179 ps**

CHECK Forgot the 2? You'd get double the frequency. N must be odd, otherwise the ring just latches.

Q10 · Dynamic power (Type 7)

$V_{DD} = 3$ V, $C_{load} = 50$ fF, clock $f = 1$ GHz (one full 0→1→0 cycle per period). Find the dynamic power, and the energy burned in the PMOS each cycle.

THEORY B9: $E = CV_{DD}^2$ per cycle; half burned in each transistor.

SOLVE

1. $E = 50\ \text{fF} \times 3^2 = 450\ \text{fJ} = 0.45\ \text{pJ}$ per cycle.
2. $P = E \times f = 0.45\ \text{pJ} \times 10^9 =$ **0.45 mW**
3. PMOS: $\frac{1}{2}CV_{DD}^2 =$ **0.225 pJ** per cycle (the NMOS burns the other 0.225 pJ).

CHECK The PMOS size does not appear: a weaker PMOS is slower, not more wasteful.

Mixed quiz question (Lectures 23 + 24 + 25)

$V_{DD} = 2.5$ V, $V_{T0,n} = 0.5$ V, $V_{T0,p} = -0.5$ V, $k_n = 300$, $k_p = 150$ µA/V², $C_{load} = 50$ fF. (a) V_{th}. (b) Exact τ_{PHL}, τ_{PLH}, τ_P. (c) How do you make the edges equal, and what happens to V_{th}?

THEORY (a) Lecture 24 V_{th} formula. (b) B5. (c) B8 rule 2 → symmetric inverter.

SOLVE

1. (a) $k_R = 2$, $\sqrt{1/k_R} = 0.7071$. $V_{th} = \dfrac{0.5 + 0.7071 \times 2.0}{1.7071} =$ **1.121 V** (left of 1.25 V: the NMOS is stronger).
2. (b) Both have $u = 2.0$ V, so the same bracket: $\frac{2(0.5)}{2.0} + \ln(4 \times 2.0/2.5 - 1) = 0.5 + \ln 2.2 = 1.2885$.
   $\tau_{PHL} = \frac{50}{300 \times 2} \times 1.2885 =$ **107 ps**; $\tau_{PLH} = \frac{50}{150 \times 2} \times 1.2885 =$ **215 ps**; $\tau_P =$ **161 ps**.
3. (c) Double $(W/L)_p$ so $k_p = k_n$. Then $\tau_{PLH} \approx 107$ ps (if C_{load} stays the same), $k_R = 1$, and V_{th} moves to $V_{DD}/2 = 1.25$ V with equal noise margins.

E · Open-book page: formulas and traps

QUESTION ASKS...	FORMULA	NOTES
C_{load}	$C_{gd,n} + C_{gd,p} + C_{db,n} + C_{db,p} + C_{int} + C_g$	plate on V_{out}; no own C_{gs}
overlap	$C_{ox}WL_D$ or $C_{GDO}W$	each transistor
junction	$WEC_j + 2(W+E)C_{jsw}$	n and p values differ
next gate	$C_{ox}[(WL)_n + (WL)_p]$	usually the biggest
quick estimate	$\tau \approx CV_{DD}/I$ or $C\Delta V/I$	Razavi style
average current	$\tau = \dfrac{C\,V_{DD}/2}{(I_{sat} + I_{lin,50\%})/2}$	sat at start, **linear at 50%**
exact delay	$\dfrac{C}{ku}\left[\dfrac{2\lvert V_T\rvert}{u} + \ln\left(\frac{4u}{V_{DD}} - 1\right)\right]$	$u = V_{DD} - \lvert V_T\rvert$
exact, $\lvert V_T\rvert \geq V_{DD}/2$	$\dfrac{CV_{DD}}{ku^2}$	saturated all the way
fall / rise time	$\dfrac{C}{ku}\left[\dfrac{2(\lvert V_T\rvert - 0.1V_{DD})}{u} + \ln\left(\frac{20u}{V_{DD}} - 1\right)\right]$	if $\lvert V_T\rvert > 0.1V_{DD}$
lone NMOS, linear	$\dfrac{C}{ku}\ln\dfrac{V_i(2u - V_f)}{V_f(2u - V_i)}$	only if $V \leq u$ throughout
lone NMOS, saturated	$\dfrac{C(V_i - V_f)}{\frac{k}{2}u^2}$	straight line
follower charging C	$\dfrac{2C}{k}\left[\dfrac{1}{V_G - V_T - V_f} - \dfrac{1}{V_G - V_T - V_i}\right]$	final value $V_G - V_T$
sizing	$\tau \propto 1/k$; equal edges: $k_p = k_n$	$(W/L)_p \approx 2.5(W/L)_n$
ring oscillator	$f = 1/(2N\tau_P)$	N odd
slow input (L26)	$\sqrt{\tau_{step}^2 + (\tau_{r,f}/2)^2}$	input rise → PHL
power	$P = CV_{DD}^2 f$; $\frac{1}{2}CV_{DD}^2$ per transistor per cycle	static ≈ 0

TOP 7 TRAPS

1. Saturation formula at the 50% point. It is **linear** there.
2. Wrong transistor: τ_{PHL} (output falls) uses the **NMOS**; τ_{PLH} uses the **PMOS**.
3. Using μC_{ox} as k. Multiply by W/L first.
4. Adding the driver's own C_{gs}, or forgetting the next gate.
5. **log** instead of **ln** on the calculator.
6. Delay (50%, input → output) mixed up with rise/fall time (10–90%, output only).
7. Forgetting the 2 in $T = 2N\tau_P$.

UNITS

fF·V/µA = ns · fF·V/mA = ps · pF·V/mA = ns · fF × V² = fJ · fJ × GHz = µW

All numerical answers were recomputed and checked against a numerical solution of $C\,dV/dt = i_p - i_n$ (square-law model, λ = 0). Tutorial answers were checked the same way (and the Tutorial 8 noise margins by sweeping the VTC). Notation follows your Lecture 23–25 guide.